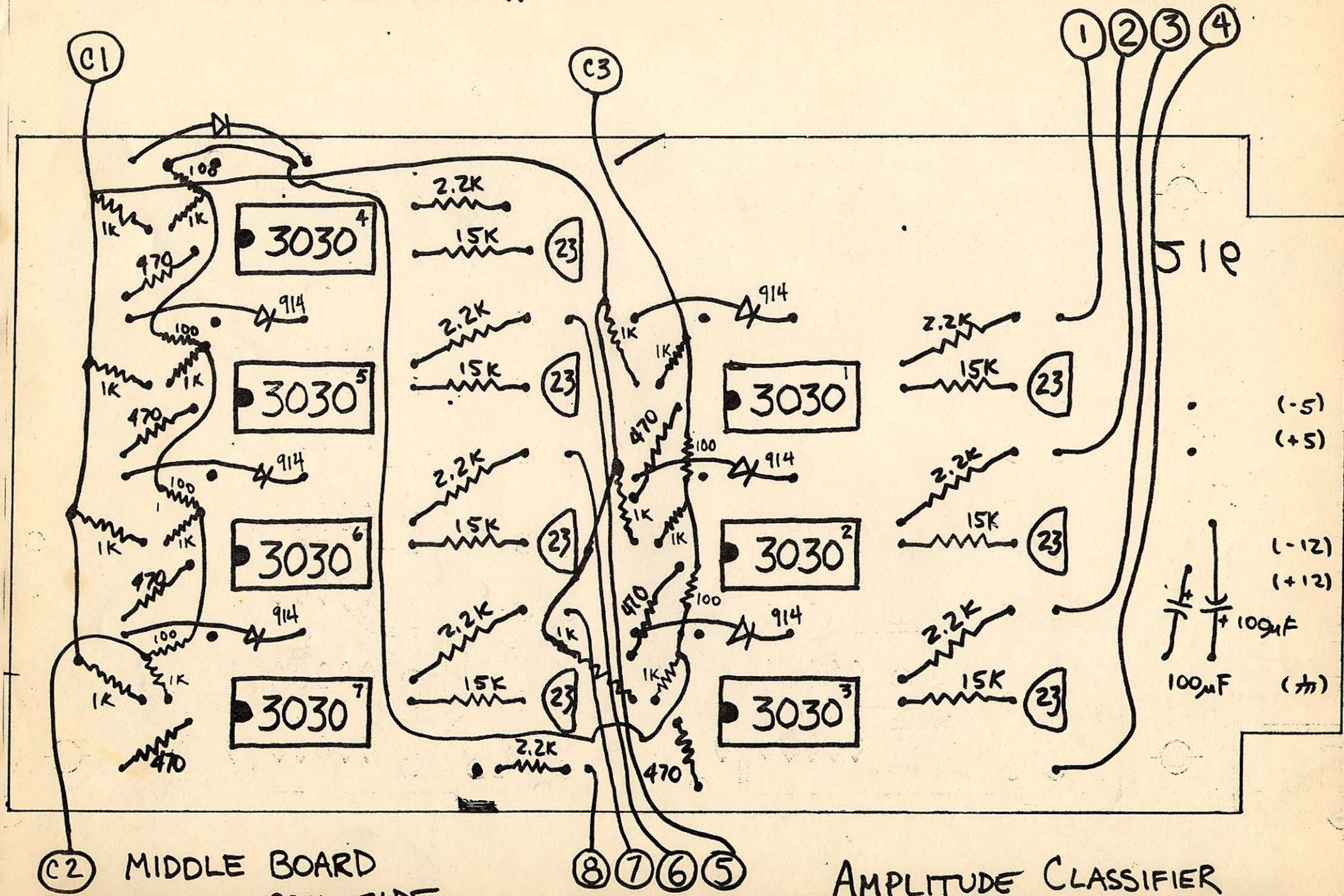
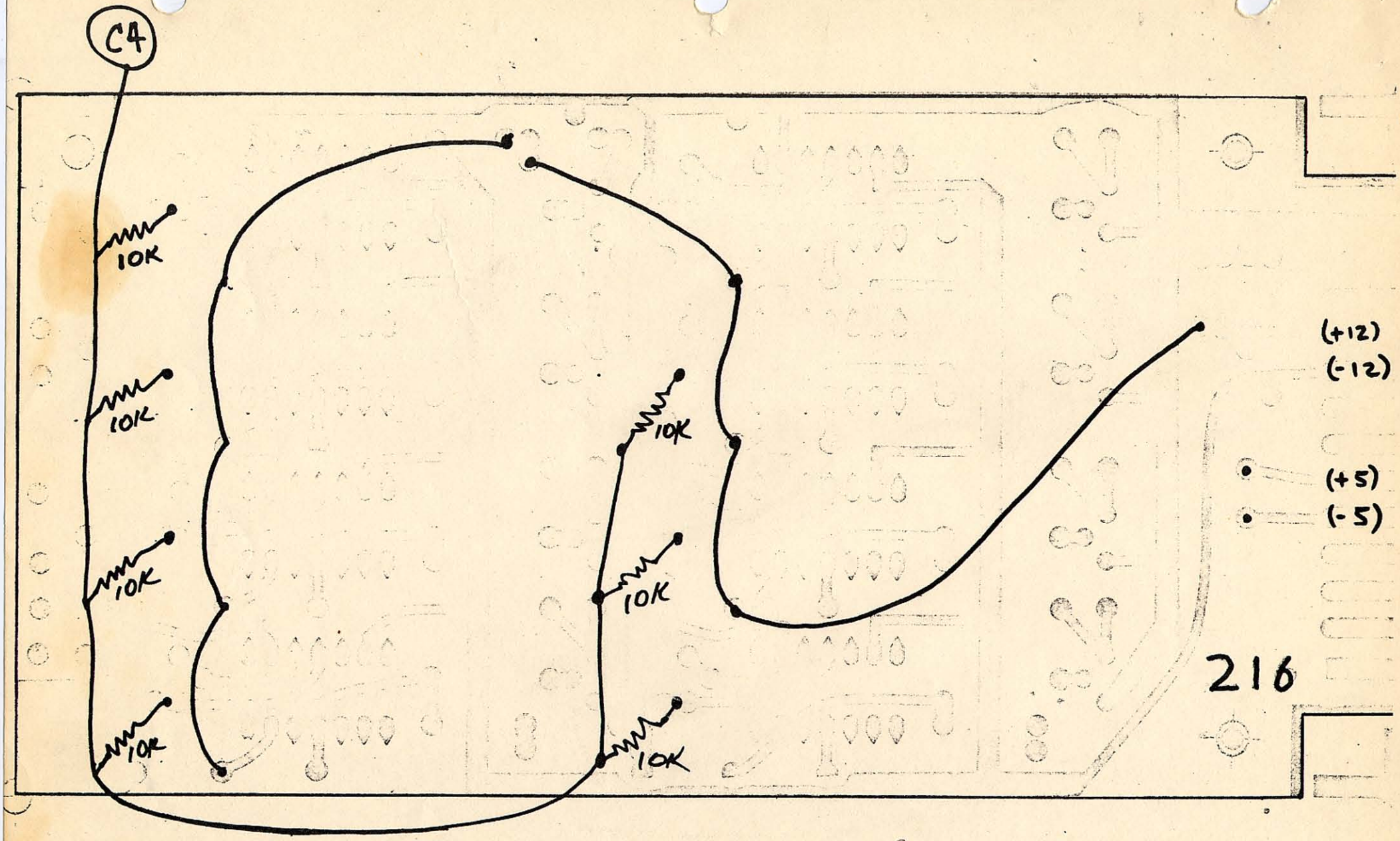
NOTE: BUS POWER TO ALL OTHER BOARDS!!



MIDDLE BOARD
COM. SIDE

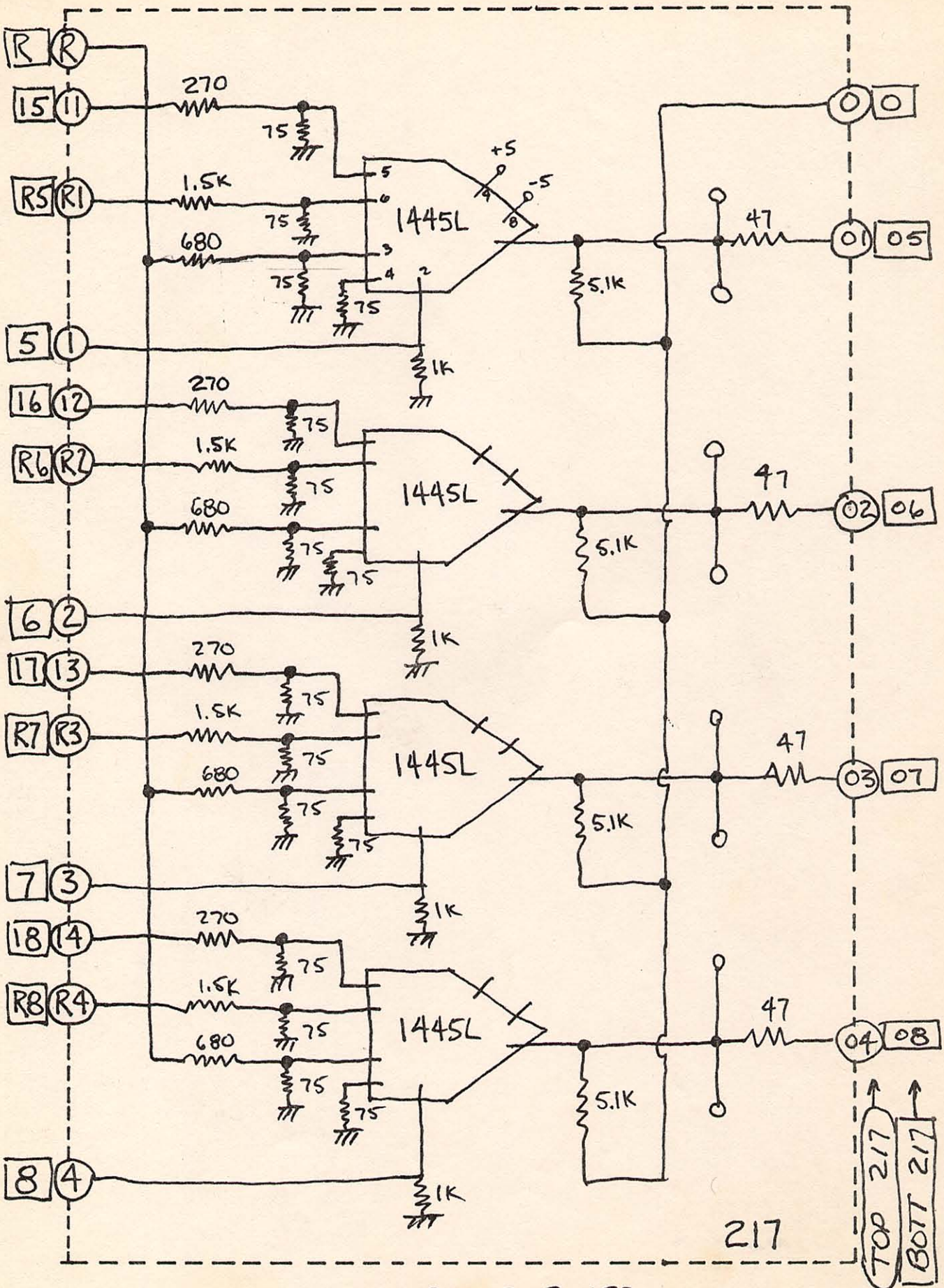
AMPLITUDE CLASSIFIER

9-75



MIDDLE BOARD
FOIL SIDE

AMPLITUDE CLASSIFIER
9-75

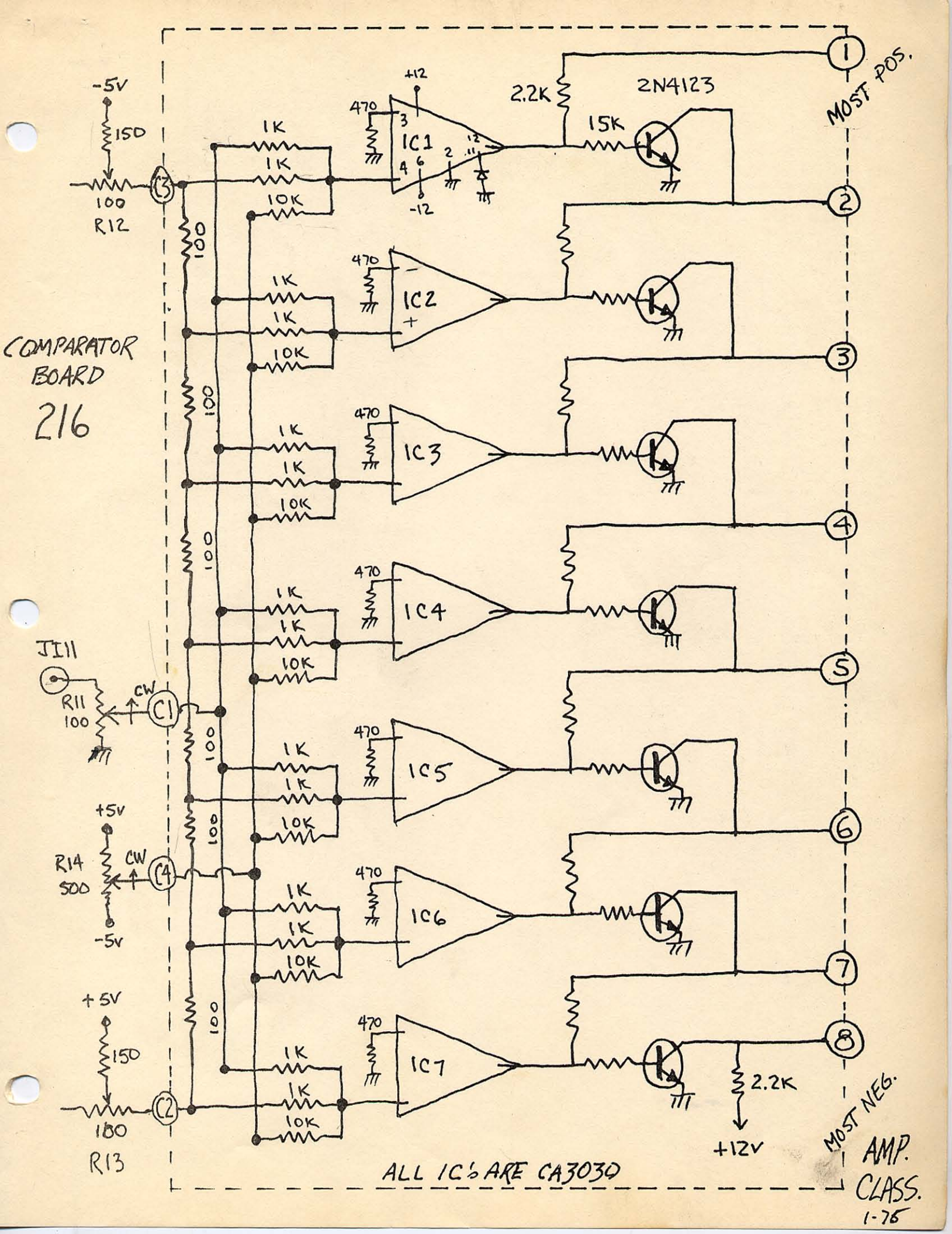


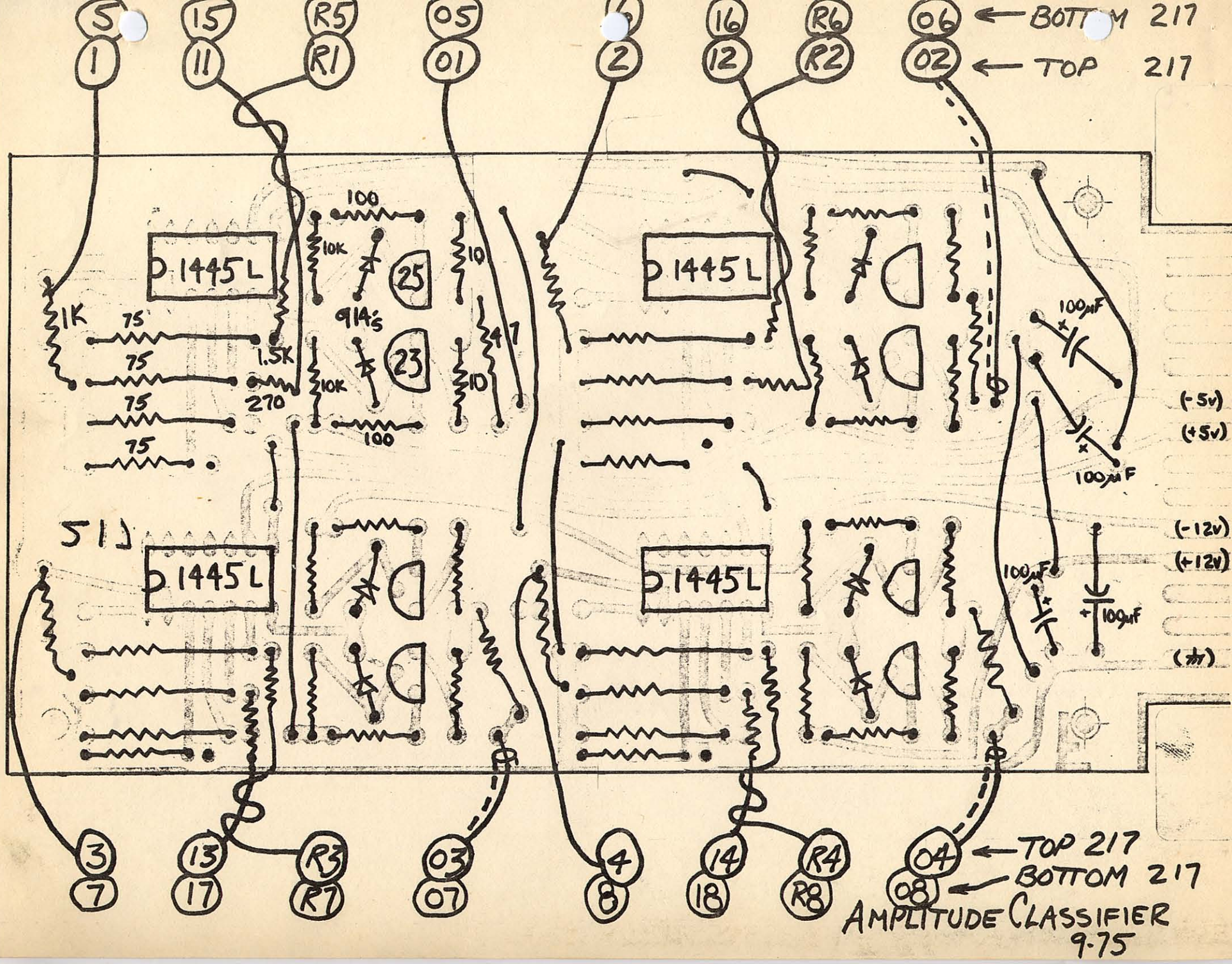
MULTIPLIER BOARD

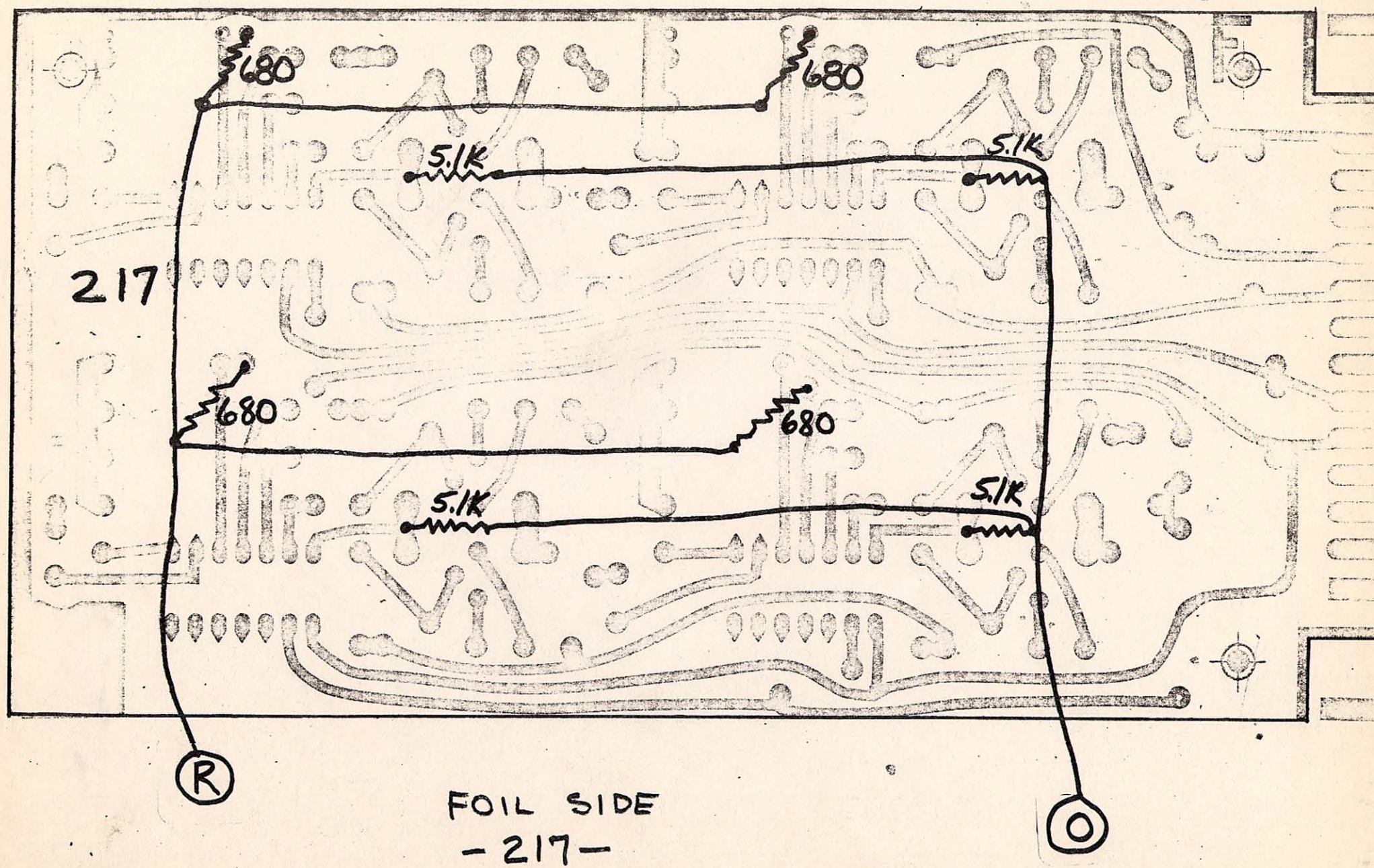
TOP OR BOTT.

AMP. CLASS,

1-76

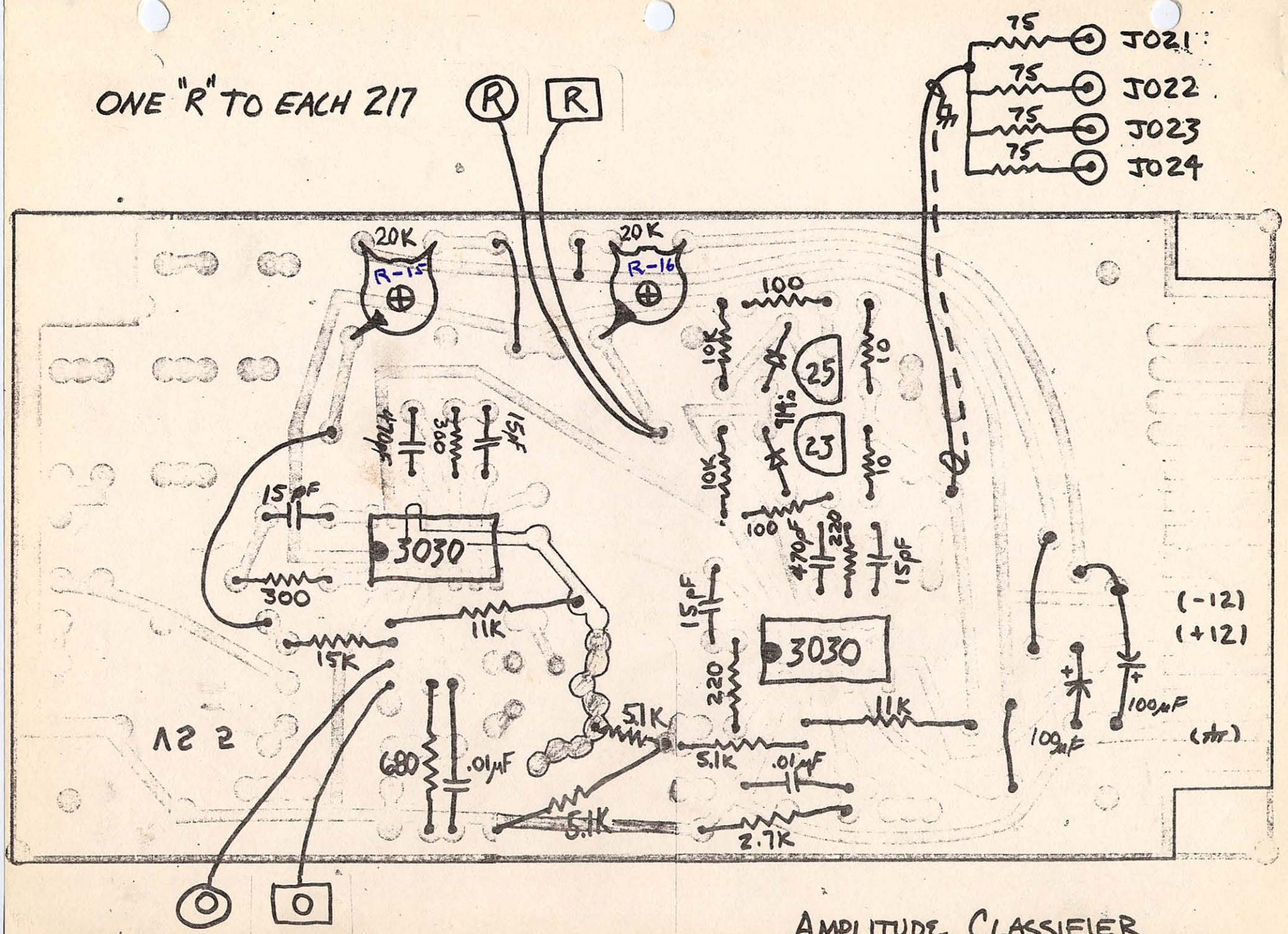

COMPARATOR BOARD 216
-5V
150
100
R12
+12
-12
470
IC1
IC2
IC3
IC4
IC5
IC6
IC7
1K
10K
100
2.2K
2N4123
15K
MOST POS.
MOST NEG.
J111
R11
100
CW
C1
C2
C3
C4
+5V
R14
500
-5V
+5V
150
100
R13
+12V
ALL IC's ARE CA3030
AMP. CLASS.
1-76

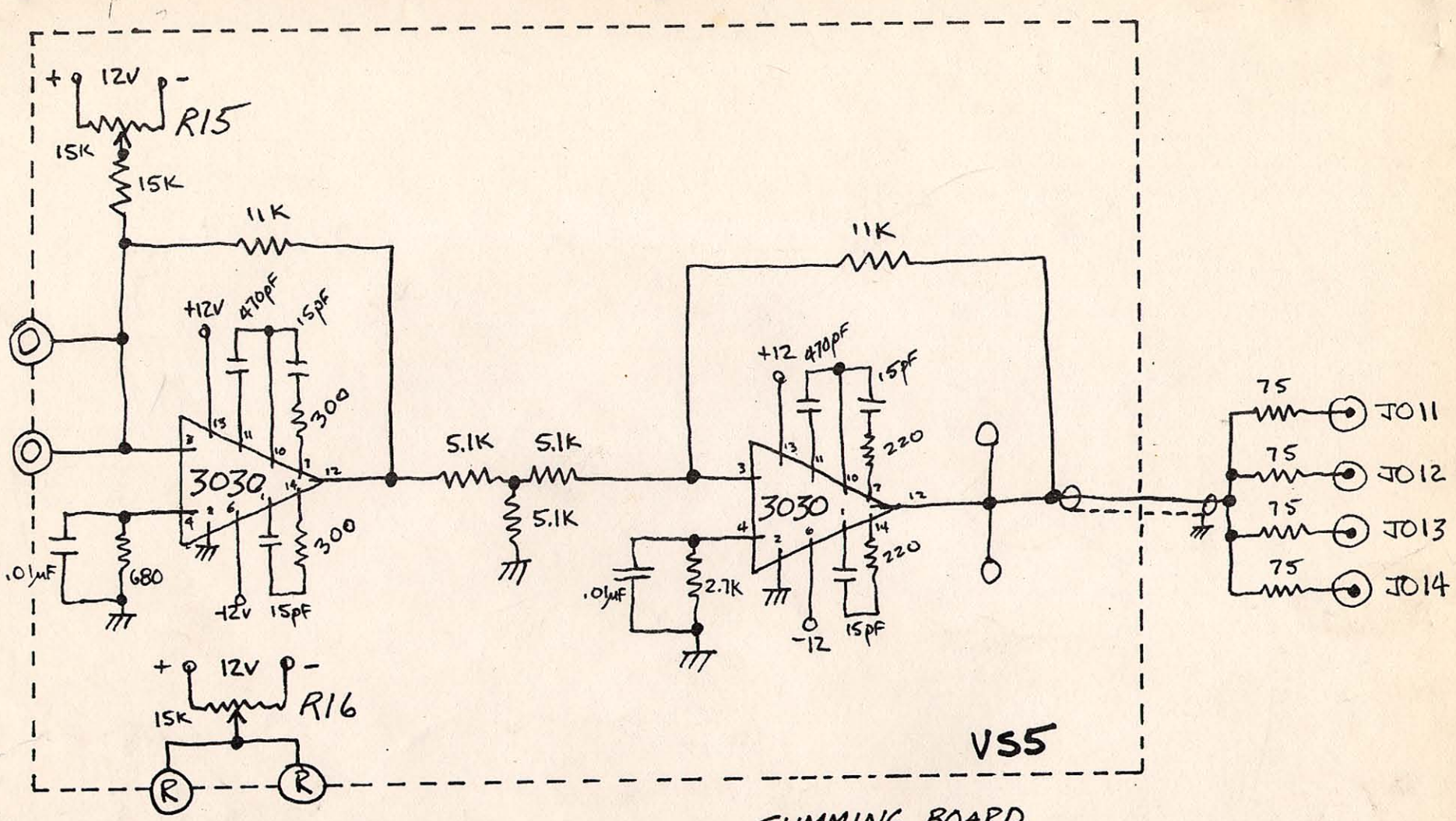

5
1
15
11
R5
R1
05
01
2
16
12
R6
R2
06
02
← BOTTOM 217
← TOP 217
100
10K
914's
25
23
10K
100
10
10
47
1K
75
75
75
75
1.5K
270
D1445L
D1445L
D1445L
D1445L
100μF
100μF
100μF
100μF
(-5v)
(+5v)
(-12v)
(+12v)
(gr)
3
7
13
17
R3
R7
03
07
4
8
14
18
R4
R8
04
08
← TOP 217
← BOTTOM 217
AMPLITUDE CLASSIFIER
9-75



FOIL SIDE
- 217 -
TOP BOARD
&
BOTTOM BOARD

AMPLITUDE CLASSIFIER
9-75


ONE "R" TO EACH 217
R
R
75
J021
75
J022
75
J023
75
J024
20K
R-15
20K
R-16
100
10K
25
10K
23
100
470pF
300
15pF
3030
300
15K
11K
15pF
220
3030
11K
5.1K
680
.01μF
5.1K
.01μF
5.1K
2.7K
(-12)
(+12)
100μF
100μF
(thr)
A2 2
O
ONE "O" TO EACH 217 BOARD
AMPLITUDE CLASSIFIER
9-75



AMP. CLASS.

1-76

18		525-1060	10 OHM, 1/4 W RES	.06	A	AC
6		525-1153	47 OHM, 1/4 W RES	.06	A	AC
36		525-1185	75 OHM, 1/4 W RES	.06	A	AC
32		525-1202	100 OHM, 1/4 W RES	.06	A	AC
2		525-1226	150 OHM, 1/4 W RES	.06	A	AC
2		525-1248	220 OHM, 1/4 W RES	.06	A	AC
8		525-1263	270 OHM, 1/4 W RES	.06	A	AC
2		525-1270	300 OHM, 1/4 W RES	.06	A	AC
7		525-1302	470 OHM, 1/4 W RES	.06	A	AC
9		525-1330	680 OHM, 1/4 W RES	.06	A	AC
22		525-1356	1K0 OHM, 1/4 W RES	.06	A	AC
8		525-1380	1.5KOHM, 1/4 W RES	.06	A	AC
8		525-1405	2.2KOHM, 1/4 W RES	.06	A	AC
1		525-1421	2.7KOHM, 1/4 W RES	.06	A	AC
11		525-1467	5.1KOHM, 1/4 W RES	.06	A	AC
25		525-1522	10K OHM, 1/4 W RES	.06	A	AC
2		525-1528	11K OHM, 1/4 W RES	.06	A	AC
7		525-1550	15K OHM, 1/4 W RES	.06	A	AC
3	10F454	101UA	100 OHM POT 1/4SFAB	1.71	N	AC
1	10F459	501UA	500 OHM POT 1/4SFT	1.71	N	AC
8	9F089	U20	10K OHM POT 1/4SFT	1.45	N	AC
2	12F9600	3389P	20K TRIM BD-MT	.65	N	AC
4	14F555	DM15-150J	15 PF,DIP-MICA CAP	.20	N	AC
2	14F592	DM15-417J	470 PF,DIP-MICA CAP	.32	N	AC
2	67F307	TDC-103Z	.01 MF,DSC-CER CAP.	.05	N	AC
12	710-1260	7-6-1000	100 MF,25VDC,ELEC.	.24	A	AC
25	553-0914	1N914B	SIL-DIODE	.19	A	AC
24		2N4123	NPN TRANS	.22	S	AC
9		2N4125	PNP TRANS	.27	S	AC
9		CA3030	DIP OP-AMP,RCA	1.32	S	AC
8		MC1445L	DIP GAIN-CONT.AMP	1.90	S	AC
21	39F1337	13-236	BNC,FM-CHS,MT.	.72	N	AC
1			CHAS,A-CFACE	8.25	DG	AC
1			VSS P-C BOARD			AC
1			216 P-C BOARD			AC
2			217 P-C BOARD			AC
1	RB-67-1-DC-M-L-9		KNOB,AMBER GAIN	1.00	R	AC
8	RB-67-0-DC-M-D-9		299 IN/MBL,D/WHT	1.00	R	AC
1	RB-67-1-DC-M-D-9		KNOB,SILVER BIAS	1.00	R	AC

SYNC GENERATOR

This module generates full NTSC color sync conforming to RS 170 EIA after trimming the master osscillator. The sync generator should stay well within the broadcast standard.

All outputs are -4v. into 75 ohm except the 14 MHZ. (JO13) which is an open collector TTL. JO13 is not used except for work with digital computers.

JI1 horizontal reset and JI2 vertical reset are not implemented in full.

The sync generator requires starting pulses which are provided by capacitors associated with JI1 and JI2. Time must be allotted after power-down before power-up (at least 30 seconds) to assure proper starting of sync generator.

Master osscillator trim:
Adjust the master osscillator frequency to 14.318180 MHZ. A convenient way to do this is to run a lead from horizontal drive and wrap it (still fully insulated) around the antennae of a TV receiver. Tune the receiver to a station on feed from a major network (in color). You will notice two vertical lines or one vertical bar drifting across the screen. Adjust the master osscillator with a long insulated screwdriver through the access hole in the front panel of module until vertical lines do not drift.

(HOR) JI1 • JO1 • JO2 • (3.58) COLOR CARRIER

(VERT) JI2 • JO3 • JO4 • (BF) BURST FLAG

JO5 • JO6 • (CS) COMPOSITE SYNC

JO7 • JO8 • (HD) HORIZONTAL DRIVE

JO9 • JO10 • (CB) COMPOSITE BLANK

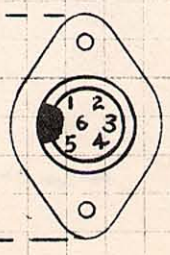
JO11 • JO12 • (VD) VERTICAL DRIVE

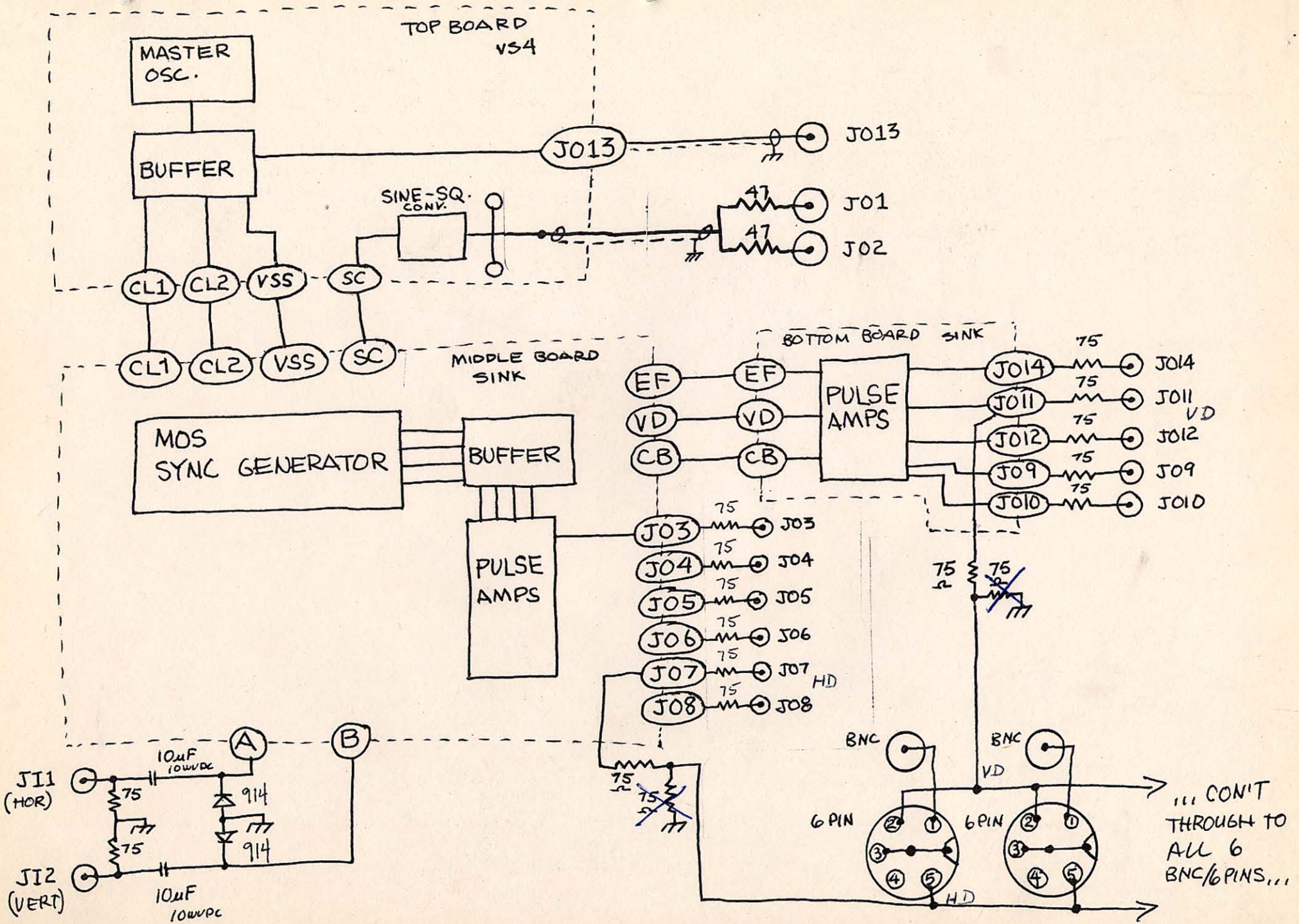
JO13 • JO14 • (JO13 – 14 MHZ. JO14 – EVEN FRAME)

SYNC GEN.

BNC

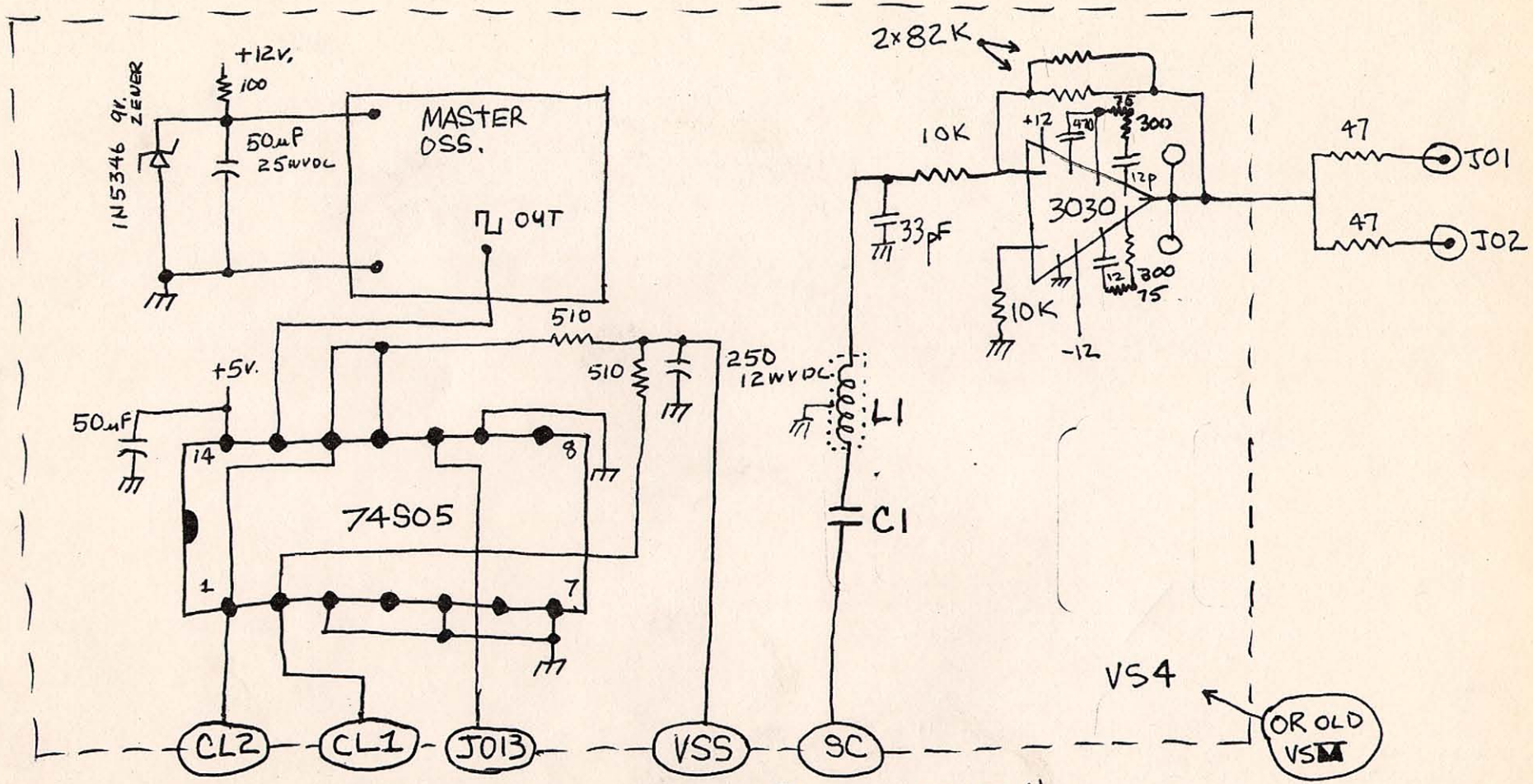
6 PIN





SYNC GENERATOR

TOP BOARD (VS4)

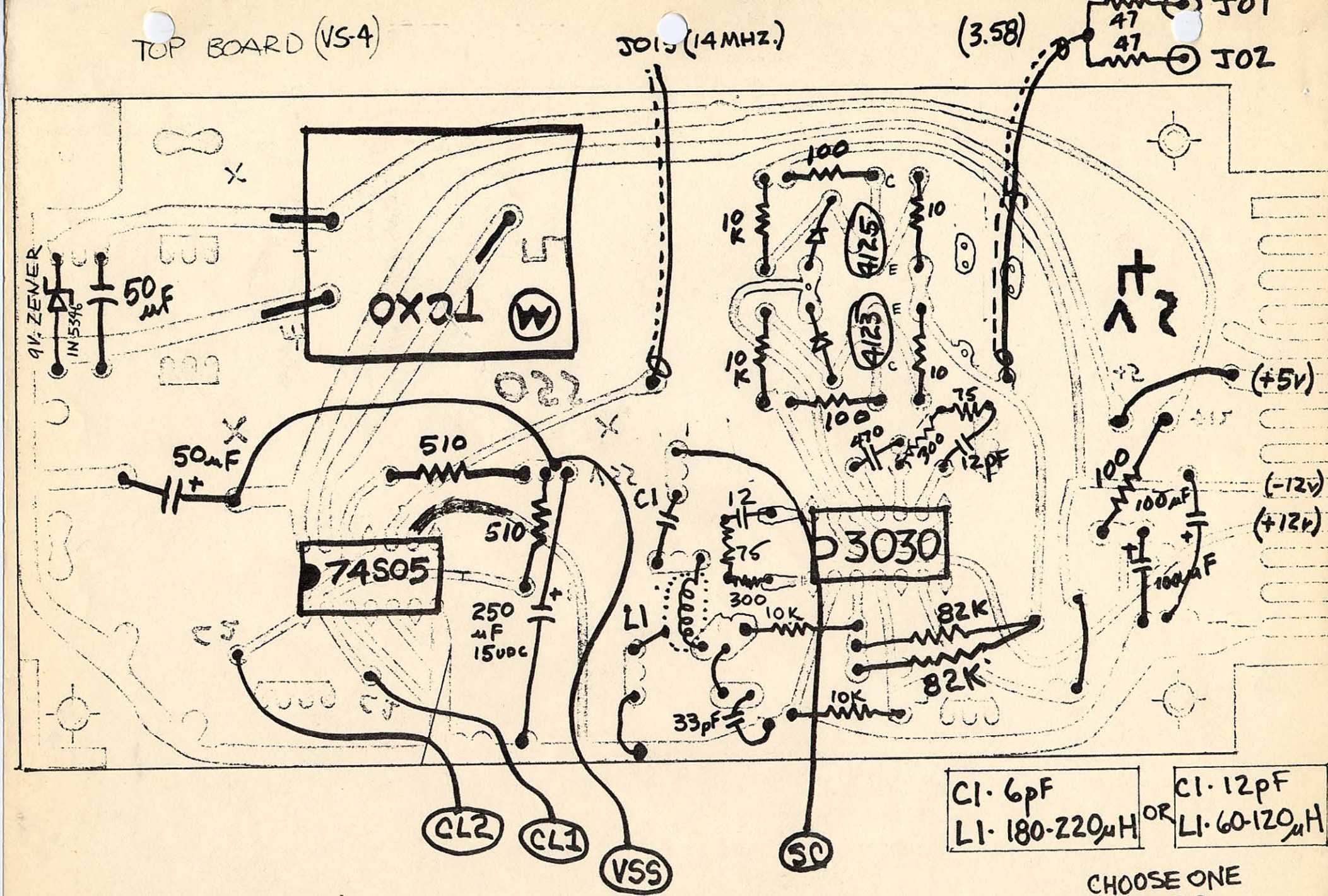


NOTE: L1 = 120-280μH SHIELDED VARIABLE CHOKE } OR L1 = 60-120μH
C1 = 6pF C1 = 12pF
MASTER OSSCILLATOR 14.318180 MHZ.
K1087A-2T1B (MICROSONICS)
DO NOT SUBSTITUTE 74S05 (FAIRCHILD)

SYNC GENERATOR

update 10-75

TOP BOARD (VS-4)



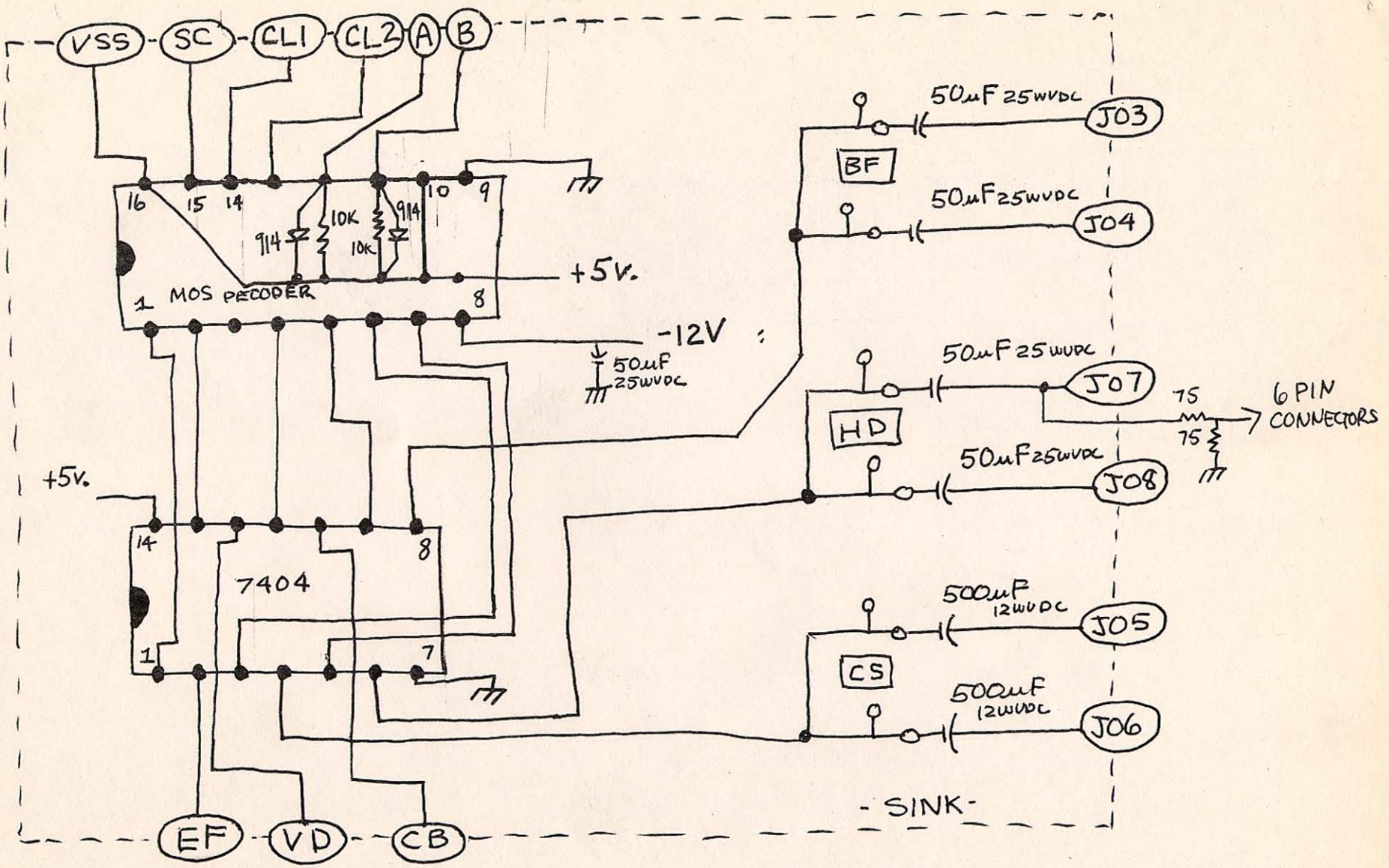
C1 · 6pF	OR	C1 · 12pF
L1 · 180-220 μH		L1 · 60-120 μH

CHOOSE ONE COMBO ONLY

SYNC GENERATOR

update 10-75

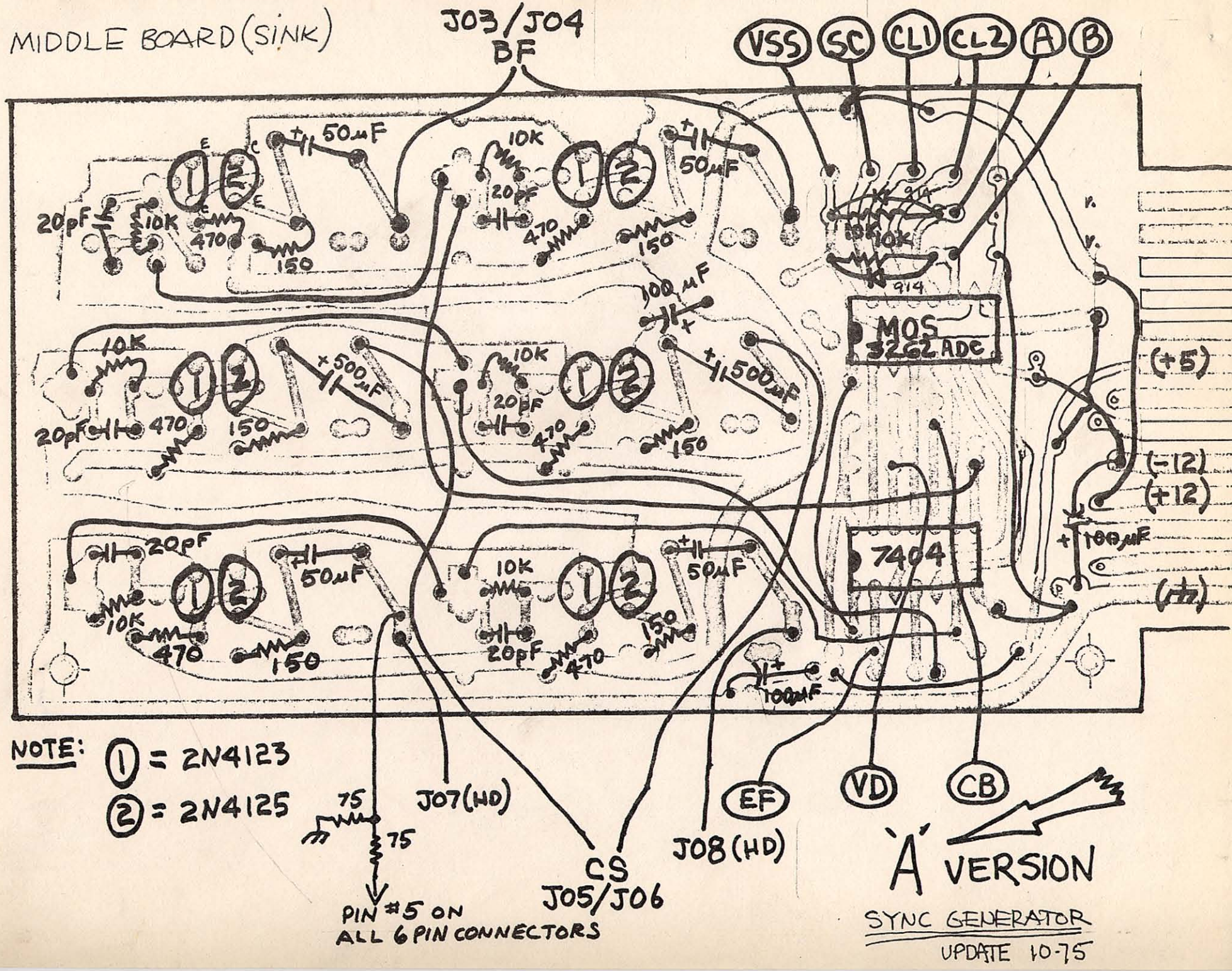
MIDDLE BOARD (SINK)

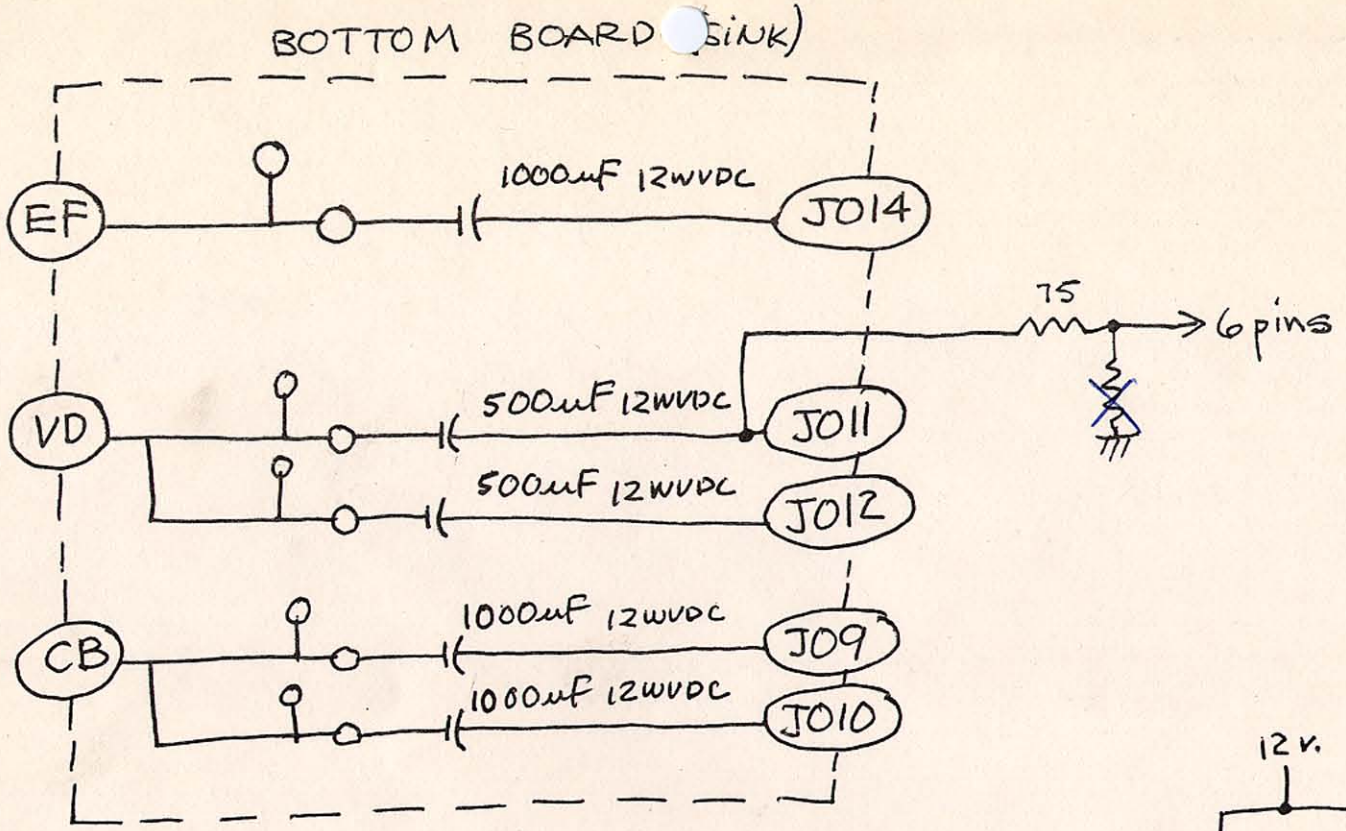


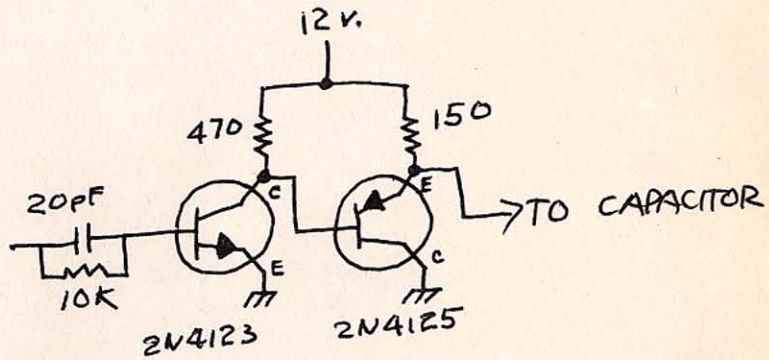
NOTE: USE 3262 ADC DECODER

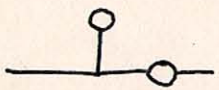
SYNC GENERATOR

UPDATE: 10-75


MIDDLE BOARD (SINK)
J03/J04
BF
VSS
SC
CL1
CL2
A
B
50µF
10K
20pF
470
150
100µF
MOS
32 CHAN ADC
7404
(+5)
(-12)
(+12)
914
NOTE:
① = 2N4123
② = 2N4125
75
J07(HD)
CS
J05/J06
J08(HD)
EF
VD
CB
PIN #5 ON
ALL 6 PIN CONNECTORS
'A' VERSION
SYNC GENERATOR
UPDATE 10-75

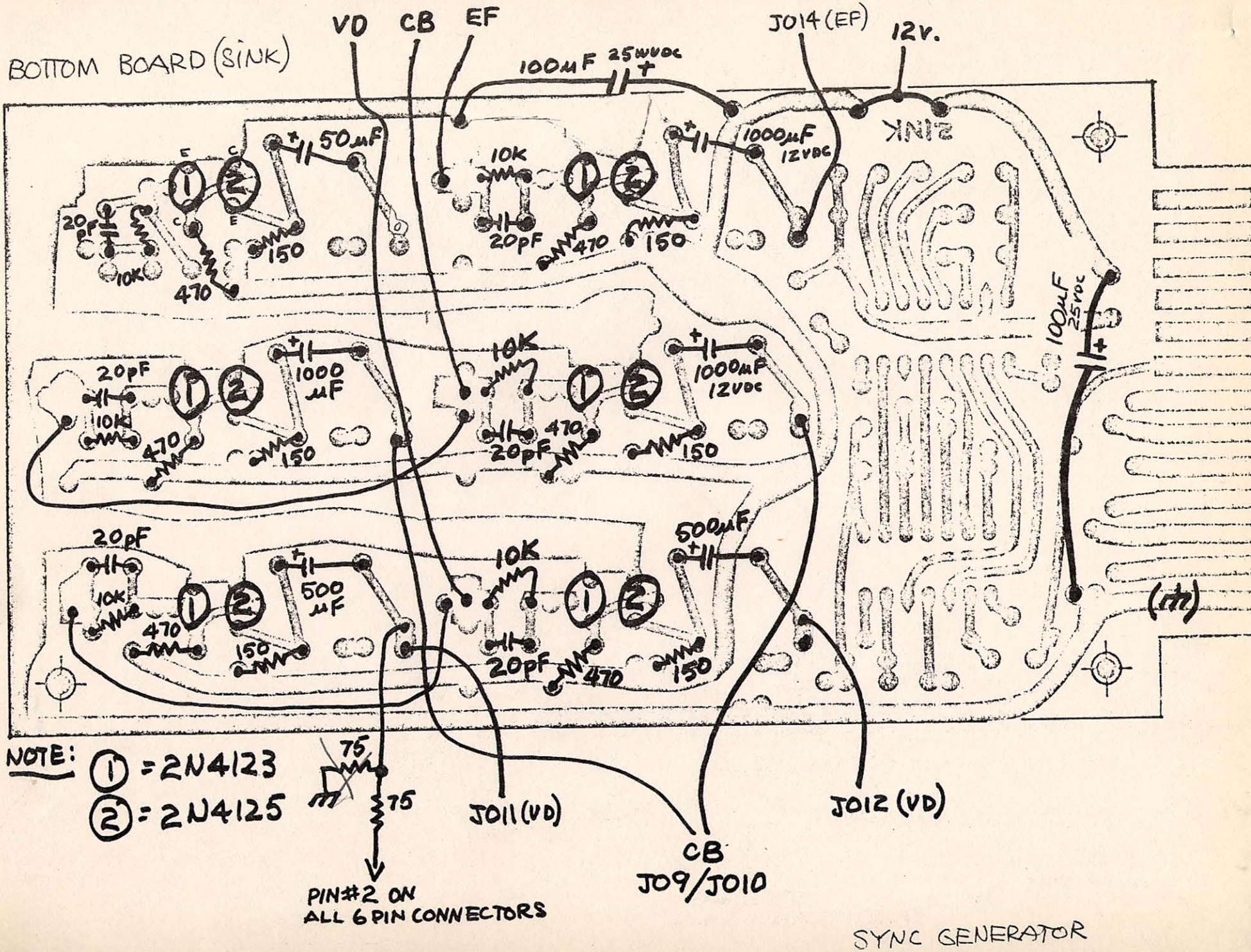




NOTE:  MEANS PULSE DRIVER AND =

SYNC GENERATOR

update 10-75


BOTTOM BOARD (SINK)
VD
CB
EF
JO14 (EP)
12v.
100μF 25WVDC
50μF
1000μF 12VDC
SINK
100μF 25VDC
20pF
10K
470
150
500μF
75
75
JO11 (VD)
JO12 (VD)
CB
JO9/JO10
PIN#2 ON
ALL 6 PIN CONNECTORS
NOTE:
① = 2N4123
② = 2N4125
SYNC GENERATOR

2		525-1060	10 OHM, 1/4 W RES	.06	A	SG
2		525-1153	47 OHM, 1/4 W RES	.06	A	SG
19		525-1185	75 OHM, 1/4 W RES	.06	A	SG
3		525-1202	100 OHM, 1/4 W RES	.06	A	SG
11		525-1226	150 OHM, 1/4 W RES	.06	A	SG
2		525-1270	300 OHM, 1/4 W RES	.06	A	SG
11		525-1302	470 OHM, 1/4 W RES	.06	A	SG
2		525-1308	510 OHM, 1/4 W RES	.06	A	SG
17		525-1522	10K OHM, 1/4 W RES	.06	A	SG
2		525-1703	82K OHM, 1/4 W RES	.06	A	SG
1	14F1267	DM5-0600	6 PF,DIP-MICA CAP	.36	N	SG
4	14F554	DM15-120J	12 PF,DIP-MICA CAP	.20	N	SG
11	14F557	DM15-200J	20 PF,DIP-MICA CAP	.18	N	SG
1	14F562	DM15-330J	33 PF,DIP-MICA CAP.	.24	N	SG
1	14F566	DM15-470J	47 PF,DIP-MICA CAP.	.24	N	SG
2	710-1251	2-G-101	10 MF,25VDC,ELEC.	.44	A	SG
6	710-1258	4-G-050	50 MF,25VDC,ELEC.	.46	A	SG
7	710-1260	7-G-1000	100 MF,25VDC,ELEC.	.24	A	SG
1	710-1218	8-D-2500	250 MF,21VDC,ELEC.	.58	A	SG
4	710-1339	L-E-500	500 MF,16VDC,ELEC.	.45	A	SG
3	710-1340	N-E-1000	1000MF,16VDC,ELEC.	.67	A	SG
1	15F2317	9056	120-280 VARI.CHOKE	3.39	N	SG
6	555-0914	1N914B	SIL-DIODE	.19	A	SG
12		2N4125	PNP TRANS	.27	S	SG
12		2N4123	NPN TRANS	.22	S	SG
1		1N5346B	9.1V ZENER 5W	2.50	N	SG
1		74805	DIP BUFF,FAIRCHILD	1.04	S	SG
1		CA3030	DIP OP-AMP,RCA	1.32	S	SG
1		3262ADC	DIP MOS DECODER	18.56	S	SG
1		7404	DIP BUF	.70	S	SG
22	39F1337	13-236	BNC,FM-CHS.MT.	.72	N	SG
1			VS4 P-CBOARD			SG
2			SINK P-CBOARD			SG
60	1-526-063-11		6-PIN FEMALE CHS MT	.90	AP	SG
1			CHASSIS,S.GEN-FACE	8.25	DG	SG
1	K1067A2T18		XTAL-14.318180MHZ	140.00	M	SG

OSCILLATOR

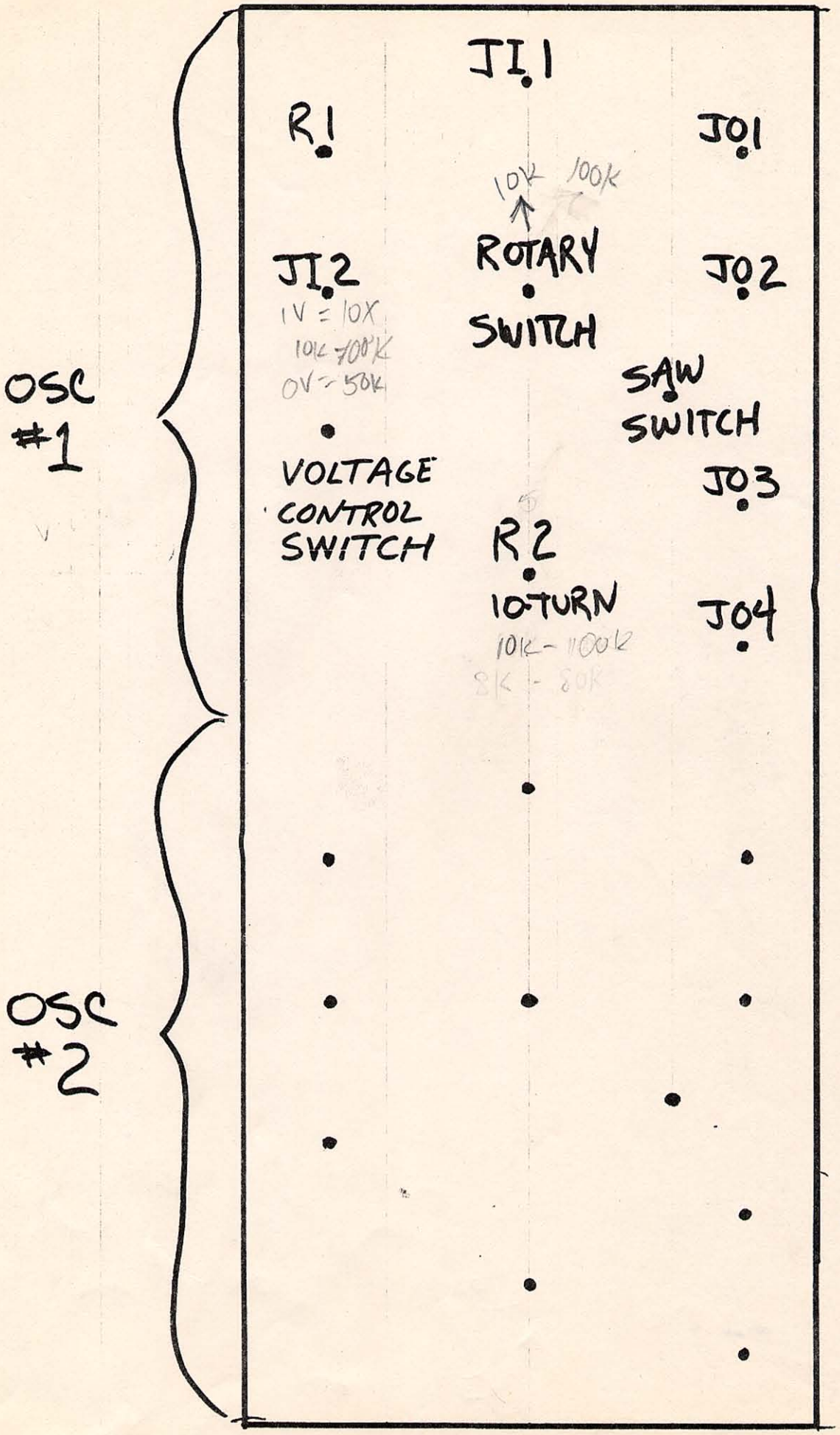
This module contains two oscillators that generates a sine wave output available at JO3, JO4 and a triangle wave output at JO1 and JO2. If the sawtooth switch is down instead of up, the triangle wave becomes a sawtooth and the sine wave becomes an "s" wave. The rotary switch sets the gross frequency range form 1/100 Hz. to ½ MHz. R2 is the continuous frequency adjustment. If the voltage control switch is up, a signal inputed to JI2 will control the frequence of the oscillator in combination with R2. With the switch down the voltage control is disabled but the oscillator is more stable. A sync. level (4 volt) signal into JI1 will trigger the oscillator to stabilize patterns.

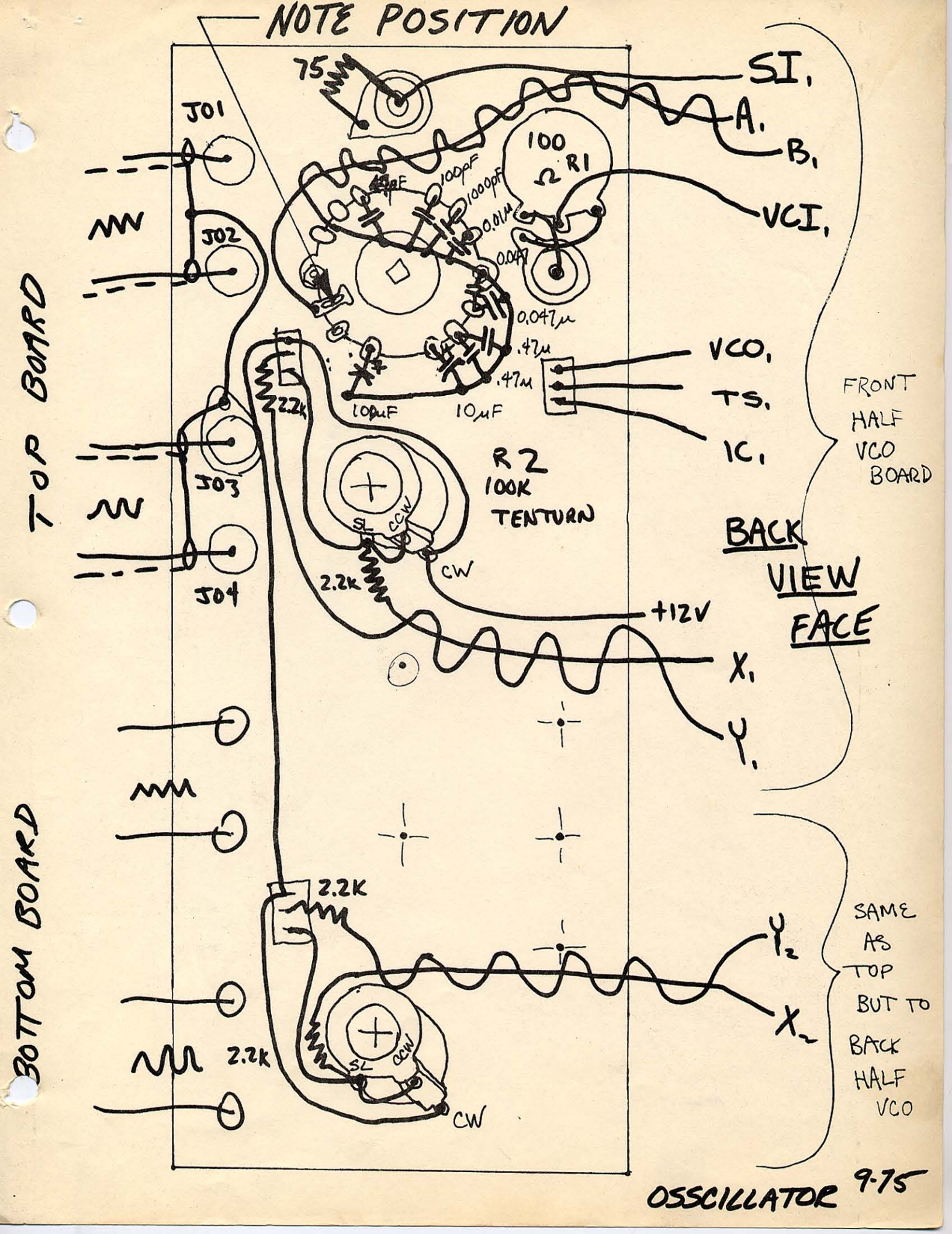
CIRCUIT DESCRIPTION

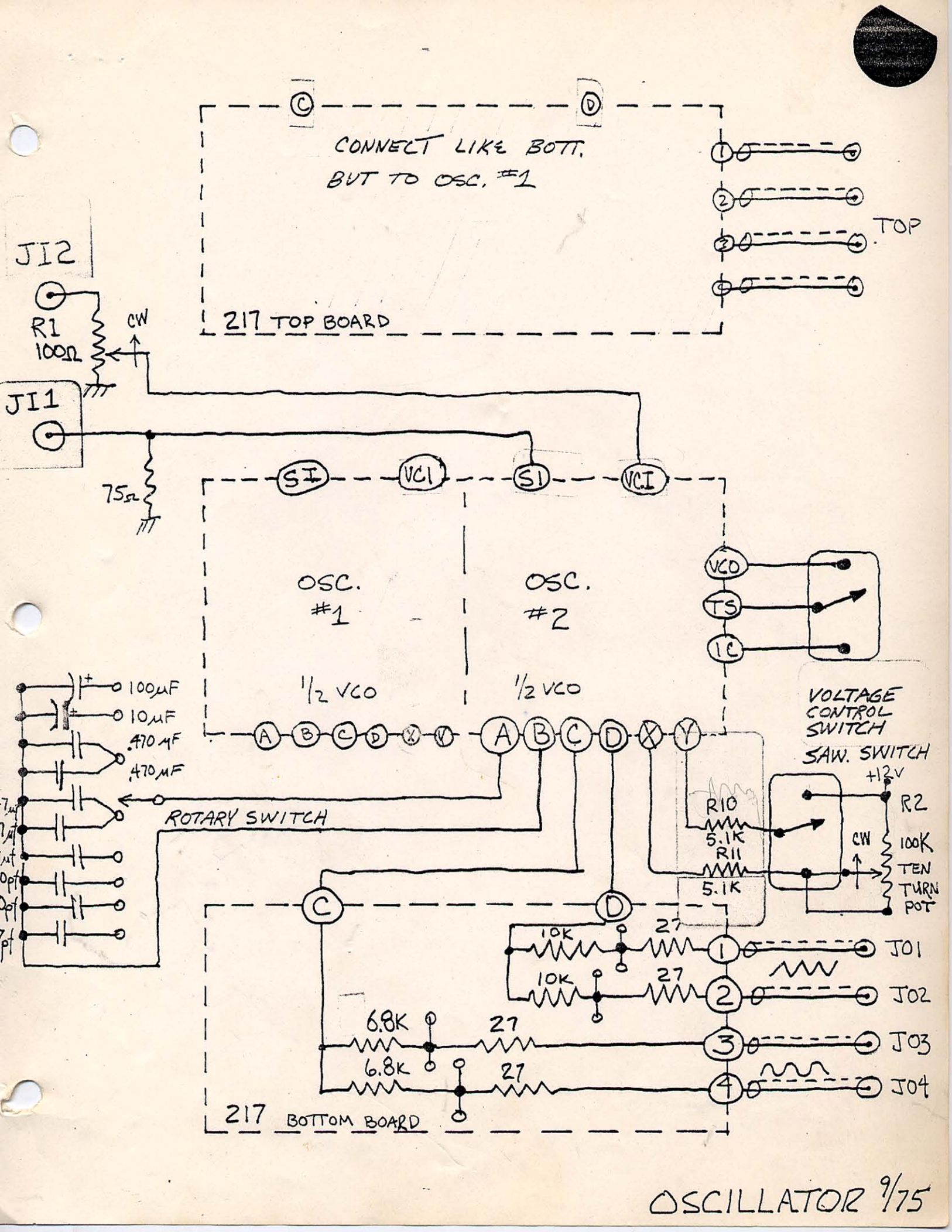
The 8038 is a complete voltage controlled oscillator whose frequency is controlled by resistance R2 and the voltage at pin 8. The 715 is a 10x amplifier and- in combination with the zener diode produces a controll voltage at pin 8.

The trimmer associated with the input of the 715 should be adjusted so that the control voltage is centered within its range with 0 volts in. To do this, input a triangle wave to the voltage control input and adjust the trimmer until the voltage that makes the maximum frequency is as positive as the voltage that quenches the oscillator is negative.

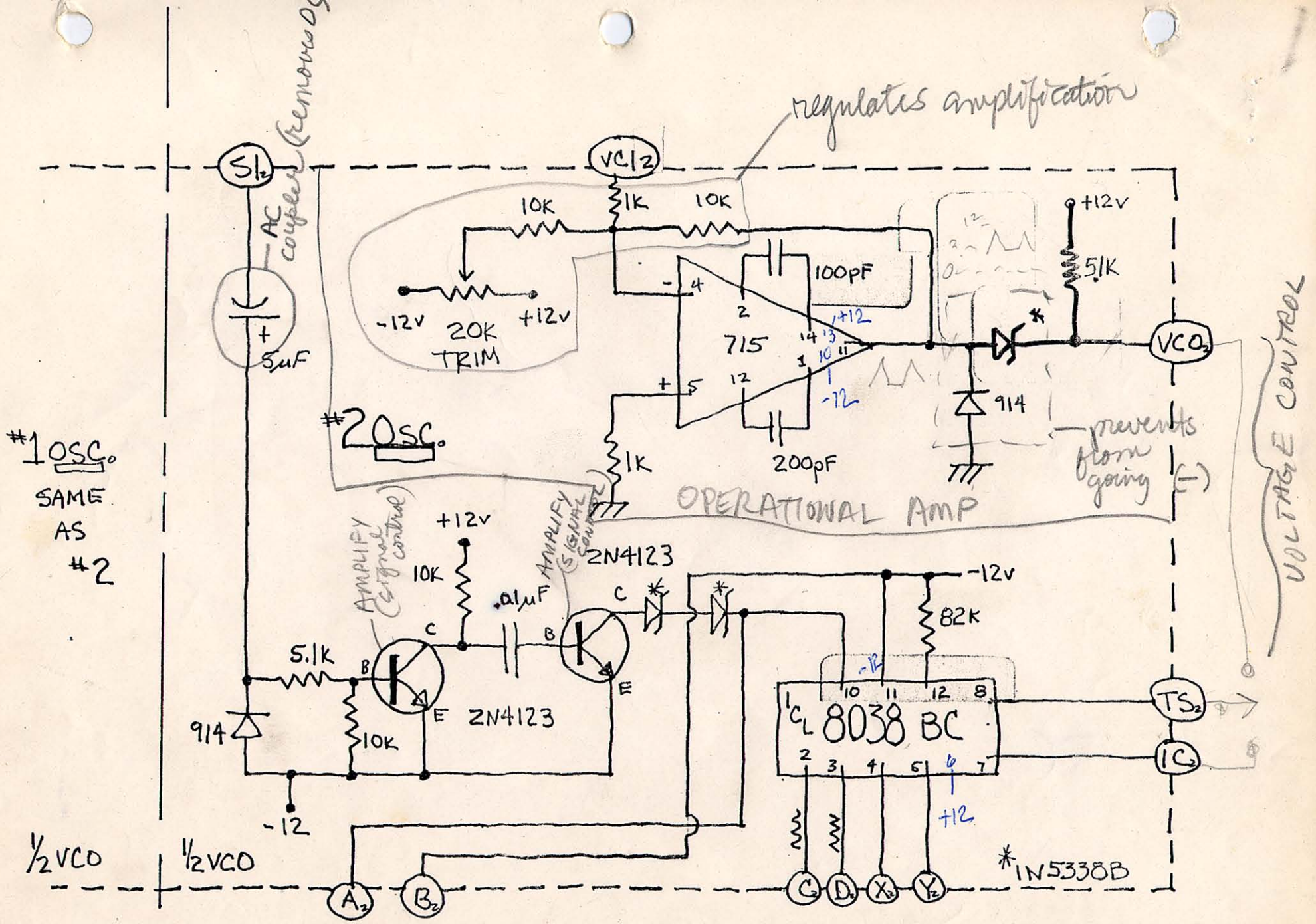
The two transistors and zeners are used to trigger the oscillator. When a fast-falling signal is presented at the sync. input this turns the first transistor off which turns the second transistor on. This clamps the oscillator to the bottom of its output wave form.


JI.1
R1
JO1
10K 100K
ROTARY
SWITCH
JI.2
1V = 10K
10K - 100K
0V - 50K
JO2
50K
SAW
SWITCH
VOLTAGE
CONTROL
SWITCH
JO3
R2
10-TURN
10K - 100K
1K - 50K
JO4
FRONT
FACE
OSC
#1
OSC
#2
OSCILLATOR
9.75


NOTE POSITION
75
SI.
A.
B.
100 Ω R1
VCI.
J01
J02
J03
J04
100pF
1000pF
0.01μ
0.047
0.047μ
.47μ
.47μ
10μF
10μF
2.2k
VCO.
TS.
IC.
FRONT HALF VCO BOARD
R2 100K TENTURN
SL
CCW
CW
2.2k
+12V
X.
Y.
BACK VIEW FACE
TOP BOARD
BOTTOM BOARD
2.2K
2.2k
SL
CCW
CW
Y2
X2
SAME AS TOP BUT TO BACK HALF VCO
OSSCILLATOR 9-75



OSCILLATOR 9/75

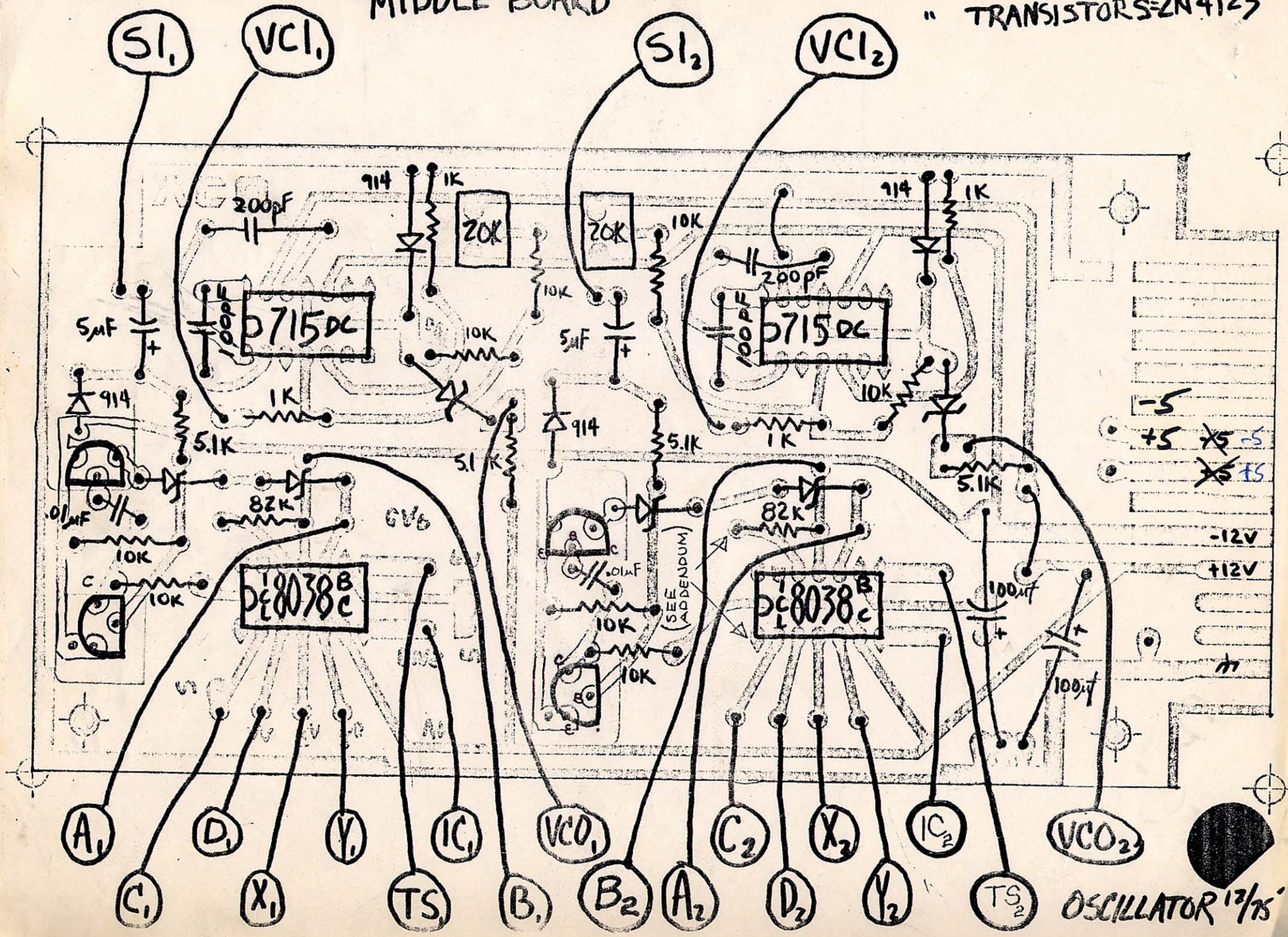


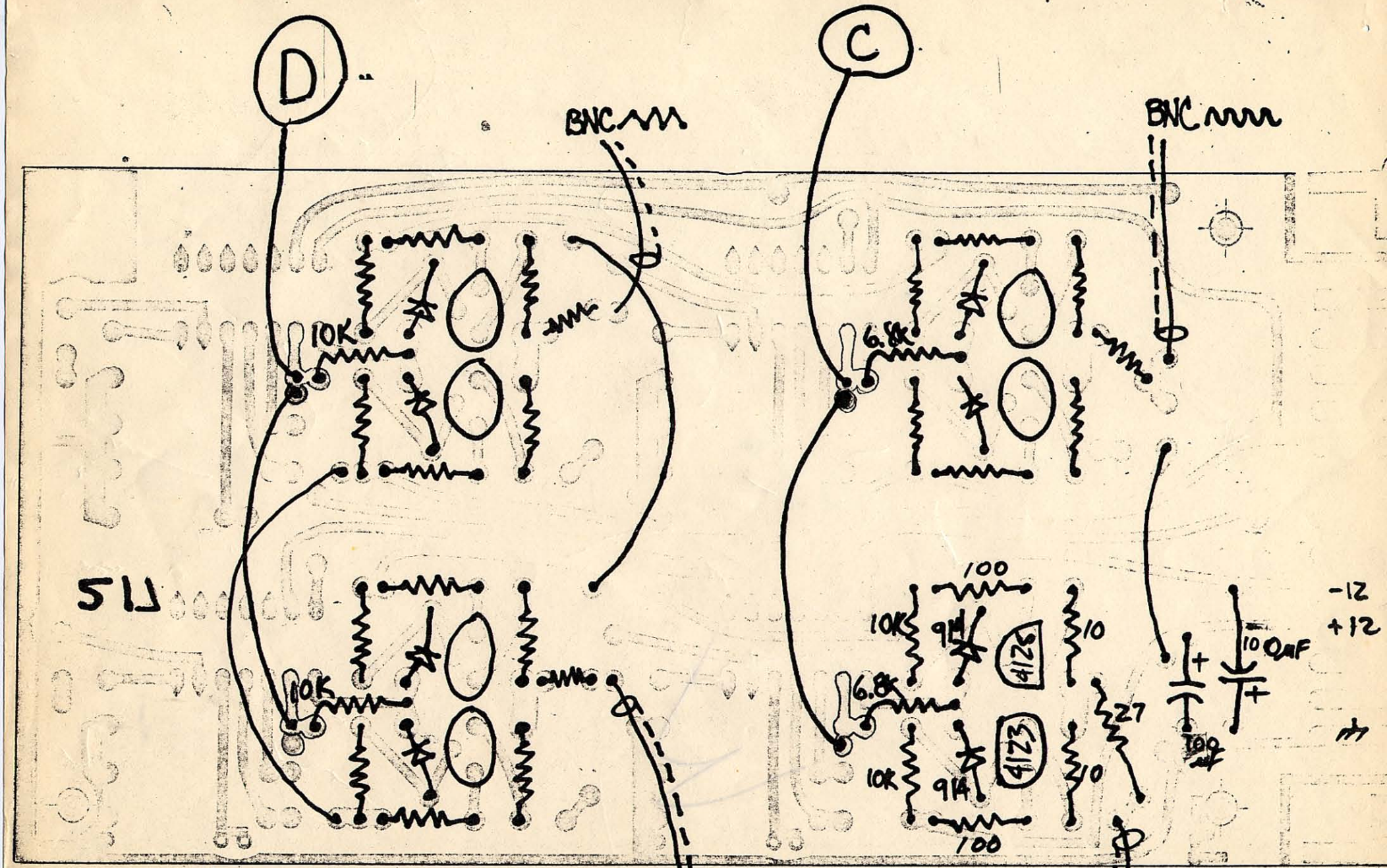
½ Middle Board VCO

Oscillator 9-75

MIDDLE BOARD

ALL ZENERS = 1N5338B
" TRANSISTORS = 2N4123





TOP & BOTTOM BOARD

OSCILLATOR

9/75

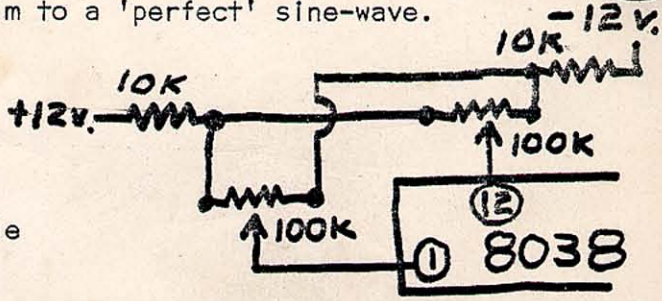
OSCILLATOR

ADDENDUM

MARCH, 1977

1.) SINE-WAVE PURITY CONTROL:

Remove 82K resistor; and, add 100K trim-pots as shown in diagram. These 100K trim-pots correct sine-wave purity. You should be able to trim to a 'perfect' sine-wave.



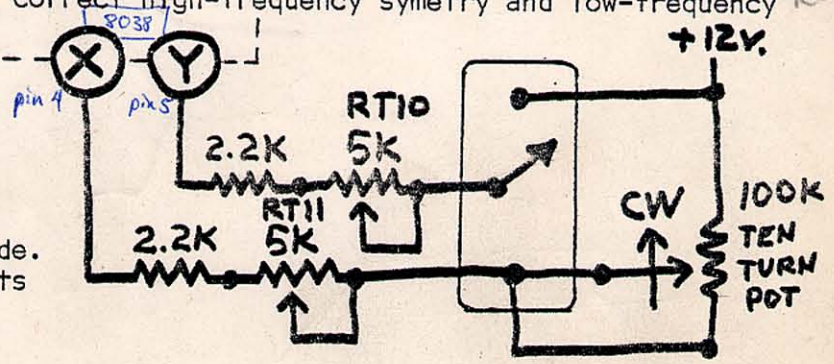
PROCEDURE-

A) Before supplying power to the module, center all trim-pots.
B) Set the oscillator at a middle freguency range, and display sine-wave on scope.
C) Tweek the trim-pots for highest amplitude possible (±1 volt) without creating any flats or peaks in the waveform; i.e. 'perfect' sine-wave.

2.) HIGH-FREQUENCY SYMETRY CONTROL:

→ SQUELCHING (TOO HIGH VOLTAGE POOR RESPONSE)

R10 and R11 maybe replaced by a series combination of 2.2K resistor and a 5K trim-pot. This series combination (RT10 and RT11) correct high-frequency symetry and low-frequency quenching of waveform; see diagram.



If both trim-pots are too large, the high-frequency end of each range will be lower than optimum.

If both trim-pots are too small, the low-frequency end in some ranges may quench, particularly in SAWTOOTH mode.

The difference between the trim-pots determines the high-frequency symetry.

PROCEDURE-

B) Turn 10-turn pot to extreme left (lowest freq.); check to make sure that no range quenches in sawtooth mode. If quenching happens in any range, tweek trim-pot to get rid of it...
C) Turn 10-turn pot to extreme right (highest-freq.); check to make sure that in a higher frequency range you still have good symetry in triangle mode. If you don't have good triangle symetry, tweek trim-pot to get it...

GO BACK AND CHECK FOR SAWTOOTH QUENCHING...

D) To maximize high-frequency in ranges, decrease both trim-pots equally and go-to-step B). If oscillator quenches at low-frequencies, back up some; i.e. increase resistance, go-to-step C). Stop.

NOTE:

These trim-pots will have to be outboarded on a perf-board and attached to card support frame of the module. Leave enough lead length on the trim-pots so it can be gotten out of the way for servicing the cards...!

Some 8038 integrated circuits appear to behave better than others; you may want to try various 8038's, choosing the best behaved ones...!

16		525-1060	10 OHM, 1/4 W RES	.06	A	OSS
8		525-1117	27 OHM, 1/4 W RES	.06	A	OSS
2		525-1165	75 OHM, 1/4 W RES	.06	A	OSS
15		525-1202	100 OHM, 1/4 W RES	.06	A	OSS
2		525-1356	1K OHM, 1/4 W RES	.06	A	OSS
2		525-1405	2.2KOHM, 1/4 W RES	.06	A	OSS
6		525-1467	5.1KOHM, 1/4 W RES	.06	A	OSS
4		525-1489	6.8KOHM, 1/4 W RES	.06	A	OSS
33		525-1522	10K OHM, 1/4 W RES	.06	A	OSS
2		525-1703	82K OHM, 1/4 W RES	.06	A	OSS
2	10F454	101UA	100 OHM POT 1/4SFAB	1.71	N	OSS
2	12F9800	3389P	20K TRIM BD-MT	.65	N	OSS
2	9F1660		100KOHM 10-TURN POT	5.30	N	OSS
2	652-0019		47PF, POLY CAP	.13	A	OSS
2	652-0026		100PF, POLY CAP	.13	A	OSS
4	652-0050		1000PF, POLY CAP	.13	A	OSS
2	652-0110		.01 MFD, POLY CAP	.13	A	OSS
4	652-0900		.047 MFD, POLY CAP	.13	A	OSS
4	652-0104		.47 MFD, POLY CAP	.13	A	OSS
2	852-4601		10 MFD, ELEC CAP, 35V	.48	A	OSS
2	852-5256		100 MFD, ELEC CAP, 150V	2.41	A	OSS
2	14F582	DM15-201J	200 PF, DIP-MICA CAP.	.20	N	OSS
1	710-1248	1-G-005	5 MF, 25VDC, ELEC.	.42	A	OSS
6	710-1260	7-G-1000	100 MF, 25VDC, ELEC.	.24	A	OSS
20	555-0914	1N914B	SIL-DIODE	.19	A	OSS
5		1N5338B	5.1V ZENER 5W	2.50	N	OSS
16		2N4123	NPN TRANS	.22	S	OSS
8		2N4125	PNP TRANS	.27	S	OSS
2		ICL8038CC	DIP OSC(V-CONT)	8.40	S	OSS
2		MA715	DIP SINK	8.25	S	OSS
2			217 P-C BOARD			OSS
1			VCO P-C BOARDS			OSS
2	22F423	1402	SW. 8-POS.	3.00	N	OSS
4	652-0507	205N	SW. SPDT	2.13	A	OSS
12	39F1337	13-236	BNC, PM-CHS. MT.	.72	N	OSS
2	~~RG-67-1-CC-M-L-9~~	~~299 1N/MBL, L6-9/WHT~~		1.00	R	~~OSS~~
2	~~RG-67-1-DC-M-L-9~~	~~3501GRY, L6-9/BL~~		1.00	R	~~OSS~~
1			CHASSIS, OSC-FACE	8.25	DG	OSS
2	61F1175	KD-1250A-1/4"	ALUM. KNOB	1.75	N	OSS
2	12F6045	DFA-N	COUNTING DIAL	4.75	N	OSS
2	12F9800	3389P	5K TRIM. BD. MT.	.65	N	OSS
2	12F9800	3389P	100K TRIM. BD. MT.	.65	N	OSS

COMPLETE PARTS LIST
for
Classical I-P

QUANTITY	SUPPLIER'S STOCK NUMBER	MANUFACTURER'S TYPE NUMBER	PART DISCRIPTION	UNIT PRICE	QUANTITY PRICE	CODE/SUPPLIER
**** NEWARK ELECTRONICS ****						
5(PKG100)	67F4065	8325	3/4" SPACER	11.00	55.00	
2(PKG100)	67F4067	8327	1" SPACER	14.00	24000	
1	29F263	1900	BOLT CUTTERS(CRIMP)	4.37	4.37	
1000 FT	36F1800WM	RG 59/U	CO-AXIAL CABLE	81.28	81.28	N
100	39F029	68175	BNC,M-CABLE MT KRMP	.79	.79	N
500 FT	36F058WF	RG 174/U	CO-AXIAL CABLE	25.25	25.25	N
1	36F650WA1	8529	WIRE,SOLID-20AWG BRN	2.19	2.19	N
1	36F650WA2	8529	WIRE,SOLID-20AWG RED	2.19	2.19	N
1	36F650WA3	8529	WIRE,SOLID-20AWG ORG	2.19	2.19	N
1	36F650WA4	8529	WIRE,SOLID-20AWG YEL	2.19	2.19	N
1	36F650WA5	8529	WIRE,SOLID-20AWG GRN	2.19	2.19	N
1	36F650WA6	8529	WIRE,SOLID-20AWG BLU	2.19	2.19	N
1	36F650WA7	8529	WIRE,SOLID-20AWG VIO	2.19	2.19	N
1	36F650WA8	8529	WIRE,SOLID-20AWG GRY	2.19	2.19	N
1	36F650WA9	8529	WIRE,SOLID-20AWG WHT	2.19	2.19	N
1	36F650WA10	8529	WIRE,SOLID-20AWG BLK	2.19	2.19	N
1(100FT)	36F590WA1	8500	WIRE,STR-16AWG BRN	2.96	2.96	N
1(100FT)	36F590WA2	8500	WIRE,STR-16AWG RED	2.96	2.96	N
1(100FT)	36F590WA3	8500	WIRE,STR-16AWG ORG	2.96	2.96	N
1(100FT)	36F590WA4	8500	WIRE,STR-16AWG YEL	2.96	2.96	N
1(100FT)	36F590WA5	8500	WIRE,STR-16AWG GRN	2.96	2.96	N
1(100FT)	36F590WA6	8500	WIRE,STR-16AWG BLU	2.96	2.96	N
1(100FT)	36F590WA7	8500	WIRE,STR-16AWG VIO	2.96	2.96	N
1(100FT)	36F590WA8	8500	WIRE,STR-16AWG GRY	2.96	2.96	N
1(100FT)	36F590WA9	8500	WIRE,STR-16AWG WHT	2.96	2.96	N
1(100FT)	36F590WA10	8500	WIRE,STR-16AWG BLK	2.96	2.96	N
2	36F1260	17656	POWER CORDS 14 AWG.	2.73	5.46	N
1	60F2408	15-15	FUZED AC OUTLET BOX	12.85	12.85	N
1	27F692	31-3001	FUZE,1 A.-3AG,SB	1.50	1.50	N
8(BOX100)	31F2245		INTER.LOCKWASH#4	.95	7.60	N
4(BOX100)	30F698		BINDHEAD4-40*1/4	1.84	7.36	N
1(BOX100)	30F699		BINDHEAD4-40*1/2	1.95	1.95	N
1	35F2521	W-TCP	SOLDERING STATION	37.00	37.00	N
1	35F2526	PTA	1/16 TIP (W-TCP)	1.00	1.00	N
1	58F587	45-121	WIRE STRIPPER	2.65	2.65	N
1	34F093	0257-4C	CUTTERS	3.15	3.15	N
				SUB-TOTAL=	400.92	

NEWARK ELECTRONICS
500 N. PULASKI RD.
CHICAGO, ILL. 60624

****NEWARK ELECTRONICS****

78	10F454	101UA	100 OHM POT 1/4SFAB	1.7	133.38	N
3	10F459	501UA	500 OHM POT 1/4SFT	1.7	5.13	N
6	9F073	U1	100 OHM POT 1/4SFT	1.45	8.70	N
1	10F588		500 OHM POT 1/4SFTB	5.02	5.02	N
3	9F078	U4	1K OHM POT 1/4SF	1.45	4.35	N
1	9F081	U6	2K OHM POT 1/4SFT	1.45	1.45	N
19	10F468	103UA	10K OHM POT 1/4SFAB	2.21	41.99	N
32	9F089	U20	10K OHM POT 1/4SFT	1.4	46.40	N
3	10F473	503UA	50K OHM POT 1/4SFAB	2.21	9.00	N
4	9F1860		100KOHM 10-TURN POT	5.3	21.20	N
3	10F683	501UC	500 TRIM,LOK,PNL-MT	3.00	9.00	N
15	10F655		50K TRIM,LOK,PNL-MT	3.00	45.00	N
28	12F9800	3389F	20K TRIM BD-MT	.6	18.20	N
4	14F1267	DM5-0600	6 PF,DIP-MICA CAP.	.36	1.44	N
22	14F1269	DM5-0800	8 PF,DIP-MICA CAP.	.36	7.92	N
7	14F554	DM15-120J	12 PF,DIP-MICA CAP.	.20	1.40	N
34	14F555	DM15-150J	15 PF,DIP-MICA CAP.	.20	6.80	N
45	14F557	DM15-200J	20 PF,DIP-MICA CAP.	.18	8.10	N
1	14F562	DM15-330J	33 PF,DIP-MICA CAP.	.24	.24	N
1	14F566	DM15-470J	47 PF,DIP-MICA CAP.	.24	.24	N
3	14F567	DM15-500J	50 PF,DIP-MICA CAP.	.19	.57	N
3	14F557	DM15-101J	100 PF,DIP-MICA CAP.	.18	.54	N
4	14F582	DM15-201J	200 PF,DIP-MICA CAP.	.20	.80	N
51	14F592	DM15-471J	470 PF,DIP-MICA CAP.	.32	16.32	N
1	19F1700	835-000	.005 MFDSC-CER CAP.	.12	.12	N
27	67F307	TDC-103Z	.01 MF,DSC-CER CAP.	.05	1.35	N
1	67F305	TDC-503M	.05 MF,DSC-CER CAP.	.37	.37	N
7	67F313	TDC-104Z	.1 MF,DSC-CER CAP.	.16	1.12	N
4	32F1956	SWD 39	39 MICROHENRY CHOKE	2.91	11.64	N
4	35F1969	SWD470	470 MICROHENRY CHOK	2.91	11.64	N
1	15F2317	9656	120-280 VARI.CHOKE	3.39	3.39	N
1		1N5346B	9.1V ZENER 5W	2.50	2.50	N
15		1N5338B	5.1V ZENER 5W	2.50	37.50	N
4	22F423	1402	SW. 6-POS.	3.00	12.00	N
1	59F1740	182090	POWER SWITCH,AMBER	3.97	3.97	N
2	27F784	344125A	110V LAMP-FUZE KNOB	4.06	8.12	N
1	39F1232	160-4-N	AC RECP.FM*2POLE-GD	1.58	1.58	N
2	39F1233	160-5-N	AC RECP.M*2POLE-GRD	2.06	4.12	N
11	39-F-255	S-310-FP	JONES FM*CHS.MT.	1.21	13.31	N
2	39F1547	111-0103-001	POST BLACK 10	.52	1.04	N
2	39F1552	111-0108-001	POST BROWN 1	.52	1.04	N
2	39F1546	111-0102-001	POST RED 2	.52	1.04	N
2	39F1550	111-0106-001	POST ORANGE 3	.52	1.04	N
2	39F1551	111-0107-001	POST YELLOW 4	.52	1.04	N
2	39F1548	111-0104-001	POST GREEN 5	.52	1.04	N
2	39F1554	111-0110-001	POST BLUE 6	.52	1.04	N
2	39F1556	111-0112-001	POST VOILET 7	.52	1.04	N
2	39F1557	111-0113-001	POST GRAY 8	.52	1.04	N
2	39F1545	111-0101-001	POST WHITE 9	.52	1.04	N
314	39F1337	13-235	BNC,FM-CHS.MT.	.72	226.08	N
7	61F1175	KD-1250A-¼"	ALUM. KNOB(TIMING)	1.75	12.25	
4	12F6045	DFA-N	COUNTING DIAL	4.75	19.00	

SUB-TOTAL: 772.95
SUB-TOTAL: 400.92

NEWARK TOTAL 1173.81

****ALLIED ELECTRONICS****

154		525-1060	10 OHM, 1/4 W RES	.06	10.78	A
10		525-1081	15 OHM, 1/4 W RES	.06	1.00	A
30		525-1117	27 OHM, 1/4 W RES	.06	3.00	A
41		525-1153	47 OHM, 1/4 W RES	.06	4.10	A
243		525-1185	75 OHM, 1/4 W RES	.06	17.01	A
235		525-1202	100 OHM, 1/4 W RES	.06	16.45	A
19		525-1226	150 OHM, 1/4 W RES	.06	1.90	A
48		525-1248	220 OHM, 1/4 W RES	.06	4.80	A
79		525-1263	270 OHM, 1/4 W RES	.06	7.00	A
21		525-1270	300 OHM, 1/4 W RES	.06	2.10	A
10		525-1276	330 OHM, 1/4 W RES	.06	1.00	A
58		525-1302	470 OHM, 1/4 W RES	.06	5.80	A
66		525-1308	510 OHM, 1/4 W RES	.06	6.60	A
27		525-1330	680 OHM, 1/4 W RES	.06	2.70	A
107		525-1356	1K0 OHM, 1/4 W RES	.06	7.49	A
10		525-1361	1.1KOHM, 1/4 W RES	.06	1.00	A
30		525-1380	1.5KOHM, 1/4 W RES	.06	3.00	A
65		525-1405	2.2KOHM, 1/4 W RES	.06	6.50	A
13		525-1421	2.7KOHM, 1/4 W RES	.06	1.30	A
10		525-3427	3K OHM, 1/4 W RES	.06	1.00	A
10		525-1433	3.3KOHM, 1/4 W RES	.06	1.00	A
10		525-1461	4.7KOHM, 1/4 W RES	.06	1.00	A
134		525-1467	5.1KOHM, 1/4 W RES	.06	9.38	A
10		525-1489	6.8KOHM, 1/4 W RES	.06	1.00	A
227		525-1522	10K OHM, 1/4 W RES	.06	15.89	A
17		525-1528	11K OHM, 1/4 W RES	.06	1.70	A
32		525-1550	15K OHM, 1/4 W RES	.06	3.30	A
10		525-1603	27K OHM, 1/4 W RES	.06	1.00	A
10		525-1650	47K OHM, 1/4 W RES	.06	1.00	A
10		525-1703	82K OHM, 1/4 W RES	.06	1.00	A
10		525-1717	100KOHM, 1/4 W RES	.06	1.00	A
10		525-1907	2.2MOHM, 1/4 W RES	.06	1.00	A
241	555-0914	1N914B	SIL-DIODE	.19	45.79	A
4	852-0019		47PF,POLY CAP	.13	.52	A
4	852-0026		100PF, POLY CAP	.13	.52	A
8	852-0050		1000PF ,POLY CAP	.13	1.04	A
4	852-0110		.01 MFD, POLY CAP	.13	.52	A
8	852-0900		.047 MFD, POLY CAP	.13	1004	A
8	852-0104		.47 MFD, POLY CAP	.13	1.04	A
4	852-4601		10 MFD,ELEC CAP,35V	.48	1.92	A
4	852-5256		100 MFD,ELEC CAP,150V	2.41	9.64	A
4	710-1248	1-G-005	5 MF,25VDC,ELEC.	.42	1.68	A
11	710-1251	2-G-101	10 MF,25VDC,ELEC.	.44	4.84	A
1	710-1253	3-G-020	20 MF,10VDC,ELEC.	.20	.20	A
2	710-1254	3-G-025	25 MF,25VDC,ELEC.	.46	.92	A
7	710-1258	4-G-050	50 MF,25VDC,ELEC.	.46	3.22	A
142	710-1260	7-G-1000	100 MF,25VDC,ELEC.	.24	34.08	A
2	710-1218	8-U-2500	250 MF,212DC,ELEC.	.58	11.16	A
4	710-1339	L-E-500	500 MF,16VDC,ELEC.	.45	1.80	A
4	710-1340	N-E-1000	1000MF,16VDC,ELEC.	.67	2.68	A
4(BOX100)	920-1803	1803	TAP SCREWS #6*3/8	1.78	7.12	A
4(BOX100)	920-1365	1365	NUTS 4-40*1/4 HEX	1.62	6.48	A
1(BOX100)	920-1397	1497	SOLDER LUGS 3/8 HOLE	2.25	2.25	A
1#SPOOL	707-5060		SOLDER,60/40*.025	6.38	6.38	A
1	707X3036	3036	36 DRAWER CABINET	10.00	10.00	A
1	548Z0240	CC40	40 BIN RACK	22.50	22.50	A
8	652-0507	205N	SW. SPDT	2.13	18.48	A

TOTAL=338.62

ALLIED ELECTRONICS
1355 SLEEPY HOLLOW RD.
ELGIN, ILL. 60120

MICROSONICS
WEYMOUTH, MASS. 02188

1	K1087A2T1B	XTAL-14.318180MHZ	140.00	140.00	M
			TOTAL	140.00	

OHM ELECTRONICS
649 VERMONT AVE.
PALANTINE, ILL. 60067

2000	1938-4	05-30-0001 IC DIP SOCKETS	.036	25,50	O
1000	1875-2L	02-04-1112 TRANS. SOCKETS	.245	24.50	O
			TOTAL	50.00	

ADVENT ELECTRONICS
7110-16 N. LINCOLN AVE
ROSEMONT, ILL. 60018

10	1-526-063-11	6-PIN FEMALE CHS MT	.90	9.00	AF
30	09-02-1152	MOLEX P-C BOARD CON.	.58	17.50	AF
300	8-50110	MOLEX INSERTS	.036	10.70	AF
* 1		CEN-4092-5 COLOR ENCODER BRD	400.00	400.00	AF
* 1	713-6140	225-2222-401 44 CONTACT CON	5.89	5.89	A
			TOTAL	437.20	

LAMBDA ELECTRONICS
2420 E. OAKTON ST.
ARLINGTON HEIGHTS, ILL.

1	LOS-Z-15	15V1,4AMP POW SUP	32.00	32.00	L
2	LOS-Z-5	5V,3AMP POWER SUP.	32.00	64.00	L
1	L-12-OV-15	NON-ADJ. OVP	8.00	8.00	L
2	L-12-OV-5	NON-ADJ. OVP	8.00	16.00	L
1	LOD-W-152	12V,3AMP POWER SUP	87.00	87.00	L
2	L-12-OV-12	NON-ADJ. OVP	8.00	16.00	L
			TOTAL	207.00	

SEMI CONDUCTOR SPECIALISTS
P.O. BOX 66125
OHARA AIRPORT
CHICAGO, ILL. 60666

53	1N270	GERM-DIODE	.22	11.66	S
144	2N4123	NPN TRANS	.27	38.88	S
90	2N4125	PNP TRANS	.27	24.30	S
70	CA3030	DIP OP-AMP, RCA	1.32	92.40	S
35	MC1445L	DIP GAIN-CONT. AMP	1.90	66.50	S
3	MC9602P	DIP MONOSTBL. VIB.	8.10	24.30	S
2	MC74121P	DIP MONO. VIB.	1.18	2.36	S
4	ICL80388C	DIP OSC(V-CONT)	8.40	33.60	S
4	MA715	DIP SINK	8.25	33.00	S
1	74S05	DIP BUFF, FAIRCHILD	1.04	.1.04	S
1	3262ADC	DIP MOS DECODER	18.56	18.56	S
1	7404	DIP BUF	.70	.70	S
			TOTAL	347.30	

NOTICE: THE SONY COLOR ENCODER BOARD (FROM ADVENT) HAS BEEN DISCONTINUED.
WHEN A REPLACMENT IS FOUND AN ADDENDUM WILL FOLLOW....
-NOTE ALSO THE 44-PIN CONNECTOR (USED W/ SONY ENCODER) NEITHER ONE SHOULD BE ORDERED!!

1142.62	NEWARK
338.62	ALLIED
140.00	MICRO.
50.00	OHM
207.00	LLAMBDA
347.00	SEMI CON.
437.00	ADVENT
198.00	D&G
+ 140.58	ROGAN
3000.82	TOTAL

BOX & P.C. BOARD DATA

for

****classical I-P****

QUANTITY/I-P BOXES	MODULE NAME	P-C BOARD NAME	QUANTITY BOARDS/BOX
1	POWER SUPPLY-1	NONE	0
1	POWER SUPPLY-2	NONE	0
1	INPUT	VS5	3
1	COMPARATOR	VS5	3
1	DIFFERENTIATOR	VS5	3
1	FUNTION GENERATOR	VS5	3
1	REFERENCE	217	3
8	ADDER-MULT.	VS1	1
		VS5	1
2	OSCILLATOR	VCO	1
		217	2
1	SYNC. GENERATOR	VS4*(vsv)**	1
		SINK	2
1	SYNC. STRIPPER	VS5	4
		DS6*(f)**	1
3	AMPL. CLASSIFIER	VS5	1
		216	1
		217	2
1	COLOR ENCODER	DS6*(f)**	1
		217	1
		VS1	1

**old name in parenthisis*

FOR CLASSICAL I-P TOTAL P-C BOARD TALLY		HOLE COUNT
VS1	9	157
VS4	1	144
VS5	27	202
DS6	2	333
216	3	290
217	14	256
VCO	2	182
SINK	2	258
VS10		213

TO ORDER P-C BOARDS @ E.D.I.

1-XEROX EDI ORDER FORM(10)COPIES
2-FILL OUT ONE COPY FOR EACH TYPE BOARD
 (A)-PUT BOARD NAME IN UPPER LEFT
 (B)-PUT QUANTITY IN UPPER RIGHT
3-TO CALCULATE PRICE:
 (A)-BASE PRICE = BASE COST($1.746)XQUANTITY
 (B)-SERVICE=ETCHING COST(*varies w/ hurry&quantity*)XQUANTITY
 (C)-MISC.CHARGES:DRILLING HOLES COSTS 1 CENT EACH.....
  -MISC.CHARGE=HOLE COST per BOARD X QUANTITY
  -**note*hole cost=(hole count)X($0.01)*
4-TOTAL COST= (3A)+(3B)+(3C)x20%

SANDIN IMAGE PROCESSER

_____BOARD

ORDER FORM

QUANTITY=_____

☒ ORDER FORM (Use one form for each type board)967 ☐ QUOTATION REQUEST ☐ SEND____FORMS #

PRICE CHART

foil wt./oz	foil sides	PLATING (dip process)	XXXP PAPER PHENOLIC THICKNESS 1/32	1/16	3/32	1/8	GIO GLASS EPOXY THICKNESS 1/32	1/16	3/32	1/8	CUSTOMER SUPPLIED Non sens.	Photo sens.	
1	1	None	.040	.044	.051	.056	.072	.087	.122	.144	.050	.000	1
		Water dip lacquer	.045	.049	.056	.061	.077	.092	.127	.149	.055	.005	2
		Silver	.048	.052	.059	.064	.080	.095	.130	.152	.058	.008	3
		Tin	.047	.051	.058	.063	.079	.094	.129	.151	.057	.007	4
1	2	None	.058	.062	.069	.075	.088	.105	.134	.164	.050	.000	5
		Water dip lacquer	.063	.067	.074	.080	.093	.110	.139	.169	.055	.005	6
		Silver	.069	.073	.080	.086	.099	.116	.145	.175	.061	.011	7
		Tin	.067	.071	.078	.084	.097	.114	.143	.173	.059	.009	8
2	1	None	.042	.046	.053	.059	.074	.090	.123	.145	.050	.000	9
		Water dip lacquer	.047	.051	.058	.064	.079	.095	.128	.150	.055	.005	10
		Silver	.050	.054	.061	.067	.083	.098	.131	.153	.058	.008	11
		Tin	.049	.053	.060	.066	.081	.097	.130	.152	.057	.007	12
2	2	None	.063	.067	.075	.081	.091	.107	.137	.169	.050	.000	13
		Water dip lacquer	.068	.072	.080	.086	.096	.112	.142	.174	.055	.005	14
		Silver	.074	.078	.086	.092	.102	.118	.148	.180	.061	.011	15
		Tin	.072	.076	.084	.090	.100	.116	.146	.178	.059	.009	16
			A	B	C	D	E	F	G	H	I	J	

CHART INSTRUCTIONS:

From left side of chart, select in order, foil weight, number of foil sides, and type of PLATING.

From top of chart, select type of base material and THICKNESS.

The figure, at intersection of PLATING and THICKNESS, is base cost per square inch.

Use letter at bottom of column and number at right of row for order number.

Enter E.D.I. order number here. F12

SERVICE SCHEDULE quantity	Price/board Delivery required 1 DAY	2 DAYS	3 DAYS	1 WEEK	2 WEEKS
1	5.00	4.50	4.05	3.65	3.28
2	3.50	3.15	2.75	2.48	2.23
3	3.00	2.70	2.43	2.19	1.86
4	2.75	2.48	2.24	2.02	1.62
5	2.50	2.25	2.03	1.83	1.38
6 to 9	2.25	2.03	1.83	1.65	1.32
10 to 25	2.00*	1.80	1.62	1.46	1.10
25 to 50	1.90*	1.71*	1.54	1.31	.98
50 to 100	1.82*	1.64*	1.48	1.18	.83
100 & Up	1.77*	1.59*	1.43	1.06	.77

* Call to confirm delivery on these quantities.

$1.746 (base cost) × _____ (quantity) = base price = $_____ A

$_____ (etching cost) × _____ (quantity) = service = $_____ B

$_____ (hole cost) × _____ (quantity) = misc. charg = $_____ C

(A+B+C) X 20% = TOTAL COST = $_____ D

We are selling service. You will be notified by phone of any failure to fill order. Payment refunded.

Be sure to enclose artwork and payment or P.O. Number.

If desired, send blank check (With limit) and we figure cost.

Sign here _____

Phone Number _____

P.O. Number _____

Customer Name & Address. Correct if necessary.

E.D.I.
2615 PARK AVE.
P.O. BOX 66
CAIRO, ILL. 62914

Phone 618 734-1694 (Office)

1	[illegible]	1/030	POWER CORDS 14 AWG.	2.73	N	MISC
1	[illegible]	15-15	FUZED AC OUTLET BOX	12.85	N	MISC
1	[illegible]	51-5001	FUZE,1 A.-3AG,SB	1.50	N	MISC
30		09-02-1152	MOLEX P-C BOARD CON.	.58	AP	MISC
500	[illegible]		MOLEX INSERTS	.036	AP	MISC
2000	1933-4	05-50-0001	IC DIP SOCKETS	.036	O	MISC
1000	1873-2L	02-04-1112	TRANS. SOCKETS	.245	O	MISC
1000 FT	[illegible]	RG 59/U	CO-AXIAL CABLE	81.28	N	MISC
100	[illegible]	00175	BNC,M-CABLE MT KRMP	.79	N	MISC
500 FT	[illegible]	RG 174/U	CO-AXIAL CABLE	25.25	N	MI>"
1	[illegible]	0529	WIRE,SOLID-20AWG BRN	2.19	N	MISC
1	[illegible]	0529	WIRE,SOLID-20AWG RED	2.19	N	MISC
1	[illegible]	0529	WIRE,SOLID-20AWG ORG	2.19	N	MISC
1	[illegible]	0529	WIRE,SOLID-20AWG YEL	2.19	N	MISC
1	[illegible]	0529	WIRE,SOLID-20AWG GRN	2.19	N	MISC
1	[illegible]	0529	WIRE,SOLID-20AWG BLU	2.19	N	MISC
1	[illegible]	0529	WIRE,SOLID-20AWG VIO	2.19	N	MISC
1	[illegible]	0529	WIRE,SOLID-20AWG GRY	2.19	N	MISC
1	[illegible]	0529	WIRE,SOLID-20AWG WHT	2.19	N	MISC
1	[illegible]	0529	WIRE,SOLID-20AWG BLK	2.19	N	MISC
1(100FT)	[illegible]	0500	WIRE,STR-16AWG BRN	2.96	N	MISC
1(100FT)	[illegible]	0500	WIRE,STR-16AWG RED	2.96	N	MISC
1(100FT)	[illegible]	0500	WIRE,STR-16AWG ORG	2.96	N	MISC
1(100FT)	[illegible]	0500	WIRE,STR-16AWG YEL	2.96	N	MISC
1(100FT)	[illegible]	0500	WIRE,STR-16AWG GRN	2.96	N	MISC
1(100FT)	[illegible]	0500	WIRE,STR-16AWG BLU	2.96	N	MISC
1(100FT)	[illegible]	0500	WIRE,STR-16AWG VIO	2.96	N	MISC
1(100FT)	[illegible]	0500	WIRE,STR-16AWG GRY	2.96	N	MISC
1(100FT)	[illegible]	0500	WIRE,STR-16AWG WHT	2.96	N	MISC
1(100FT)	[illegible]	0500	WIRE,STR-16AWG BLK	2.96	N	MISC
3(BOX100)	[illegible]		INTER.LOCKWASH#4	.95	N	MI>"
4(BOX100)	[illegible]		BINDHEAD4-40*1/4	1.84	N	MISC
1(BOX100)	[illegible]		BINDHEAD4-40*1/2	1.95	N	MISC
4(BOX100)	[illegible]-1003	1803	TAP SCREWS #6*3/8	1.78	A	MISC
4(BOX100)	[illegible]-1303	1303	NUTS 4-40*1/4 HEX	1.62	A	MISC
1(BOX100)	[illegible]-1397	1497	SOLDER LUGS 3/8 HOLE	2.25	A	MISC
1#SPOOL	[illegible]-5000		SOLDER,60/40*.025	6.38	A	MISC
1	[illegible]	W-TCP	SOLDERING STATION	37.00	N	MISC.
1	[illegible]	PTA	1/16 TIP (W-TCP)	1.00	N	MISC
1	[illegible]	45-121	WIRE STRIPPER	2.65	N	MISC
1	[illegible]	0257-4C	CUTTERS	3.15	N	MISC
1	[illegible]	5056	56 DRAWER CABINET	10.00	A	MISC
1	[illegible]	CC40	40 BIN RACK	22.50	A	MISC

I-P KNOBS

quantity	stock #		cost
32	RB-67-1-DC-M-L-9	BLACK KNOBS (GAIN) #299-INLAY/MATT BLACK LINE/WHITE -9 /WHITE @$.86each	27.52
32		INLAY/SILVER (GAIN) LINE/BLACK -9 /WHITE @$.86each	27.52
33		#120-INLAY/AMBER (GAIN) LINE/BLACK -9 /WHITE @$.88each	29.04
25	RB-67-1-DC-M-D-9	BLACK KNOBS (BIAS) #299-INLAY/MATTE BLACK DOT /WHITE -9 /WHITE @$.86each	21.50
25	RB-67-0-DC-M-D-9	BLACK KNOBS (BIAS) #299-INLAY/MATTE BLACK DOT /WHITE -9 /WHITE @$1.40each	35.00
deliver to:			TOTAL $ 140.58

mail w/ check to:

TO: ROGAN KNOBS
3455 WOODHEAD DR.
NORTHBROOK, ILL.
60062

ATTN: Barbara Post

**********mail original**********cut here**********keep a copy**********

mail w/ check to:

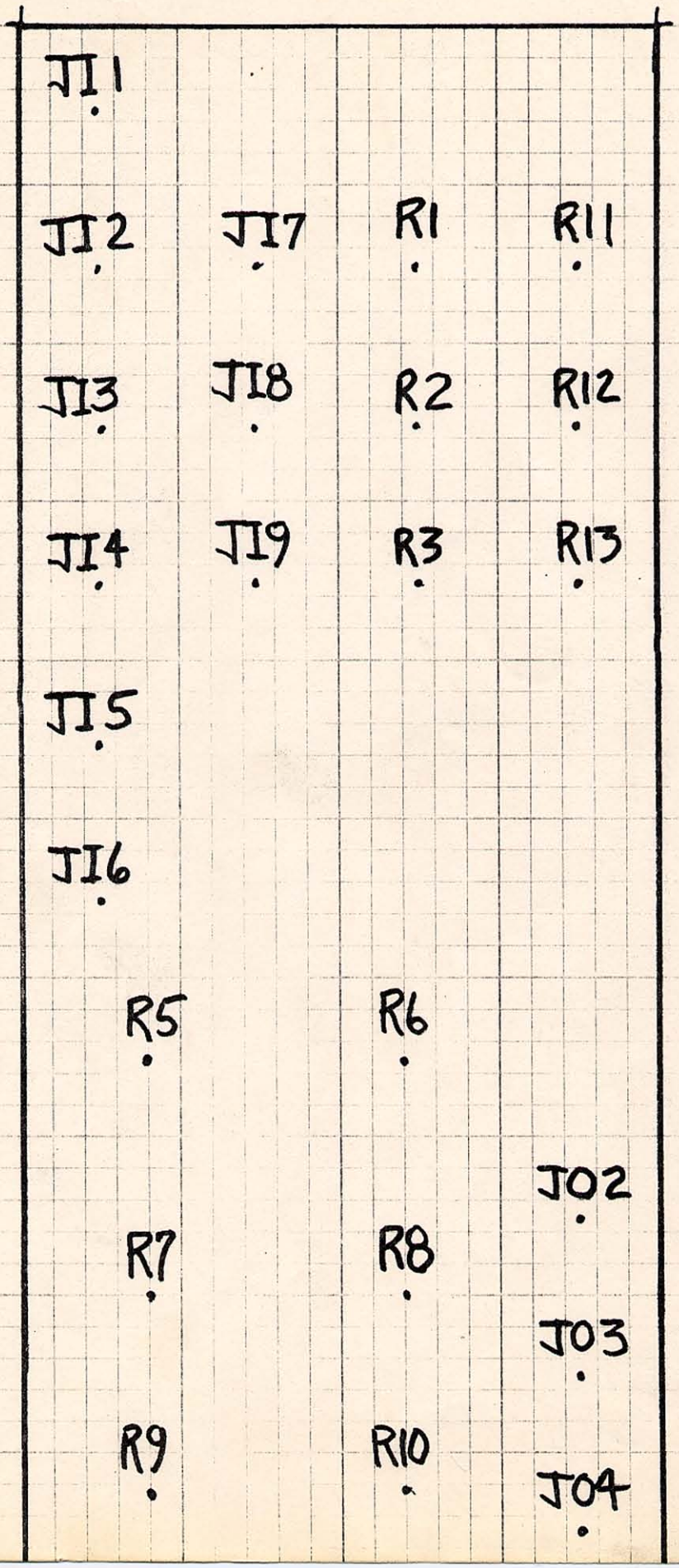
I-P BOXES
(chassis)

••• POWER SUPPLY #1	-1	deliver to:
••• POWER SUPPLY #2	-1	
••• INPUT	-1	
••• COMPARATOR	-1	name
••• DIFFERENTIATOR	-1	
••• FUNCTION GENERATOR	-1 2	address
••• REFERENCE	-1 2	
••• ADDER-MULTIPLIER	-8 9 2 mc	city
••• OSCILLATOR	-2 3	
••• SYNC. GENERATOR	-1	phone
••• SYNC. STRIPPER	-1	
••• AMPLITUDE CLASSIFIER	-3	signiture
••• COLOR ENCODER	-1	
••• BOX(blank)	-1	date

total for for classical I-P= 24

total # boxes [ ] x$8.25 = total price $ [ ]

TO: D&G ENTERPRISES
1820 S. HALSTED ST.
CHICAGO, ILL. 60608



FRONT FACE

COLOR ENCODER

9.75

COLOR ENCODER

The Color Encoder module is a N.T.S.C. standard color encoder based around the color encoder board of a Sony DXC-5000B color camera. A summary of inputs, outputs and controls follows:

JI1 COLOR CARRIER (3.58)	1v.p-p 75ohm	Supplied from Sync Generator, Sync Strip or 'House-Sync' in any TV studio...
JI2 BURST FLAG	4v.p-p 75ohm	
JI3 COMPOSITE SYNC	"	
JI4 HORIZONTAL DRIVE	"	
JI5 COMPOSITE BLANKING	"	
JI6 (reserved for future expansion)		All inputs supplied from Image Processor with associated gain controls...
JI7 RED INPUT	(+ and -).5v. 75ohm	
JI8 GREEN INPUT	"	
JI9 BLUE INPUT	"	
JI10 (not implemented yet)		

R1 through R4 are normally left at ½ gain...

JO1 LUMINANCE OUT ONLY	non-comp. 1v.p-p
JO2 N.T.S.C. OUT COMPOSITE	
JO3 "	
JO4 "	

R5 (chromance control) and R6 (hue control) act like adjustments on a color TV. Adjust using a vectorscope or adjust (visually) to a monitor tuned to a standard TV station.
R7 is adjusted to produce no chromance (color) out when there is no signal coming into the red, green, blue inputs of the encoder.
R9 (burst height), R8 (pedistal) and R10 (sync height) are adjusted to standard; if in doubt use signal from local TV station for reference (oscilliscope is necessary).

The function of the Blanking and Pedistal Board (top) is to insert blanking and pedistal to each of the monochrome input signals and route signals to the encoder board. The luminance component of the signal (red, green, and blue) is taken from the encoder and amplified by the Luminance Board (middle) and fed to the luminance output and back to the encoder. The Horizontal Clamp Board (bottom) receives horizontal sync and generates a clamping pulse which is sent to the encoder board. (pulse is positive) Adjust RT1 for a delay of 1.7 micro-seconds after horz. sync falling-edge, and adjust RT2 for a pulse length of 3 micro-seconds.

Presently, the Color Encoder Module has some 'funkiness' from a stringent point of view. It is quite adequate for all small format recording; however, in a 2" quad broadcast context, a proc-amp is recommended on its output to prevent oversaturation of chromance information.

MODIFICATIONS on the SONY ENCODER BOARD:
Remove the white delay line (DL2); it has four leads to be de-soldered...

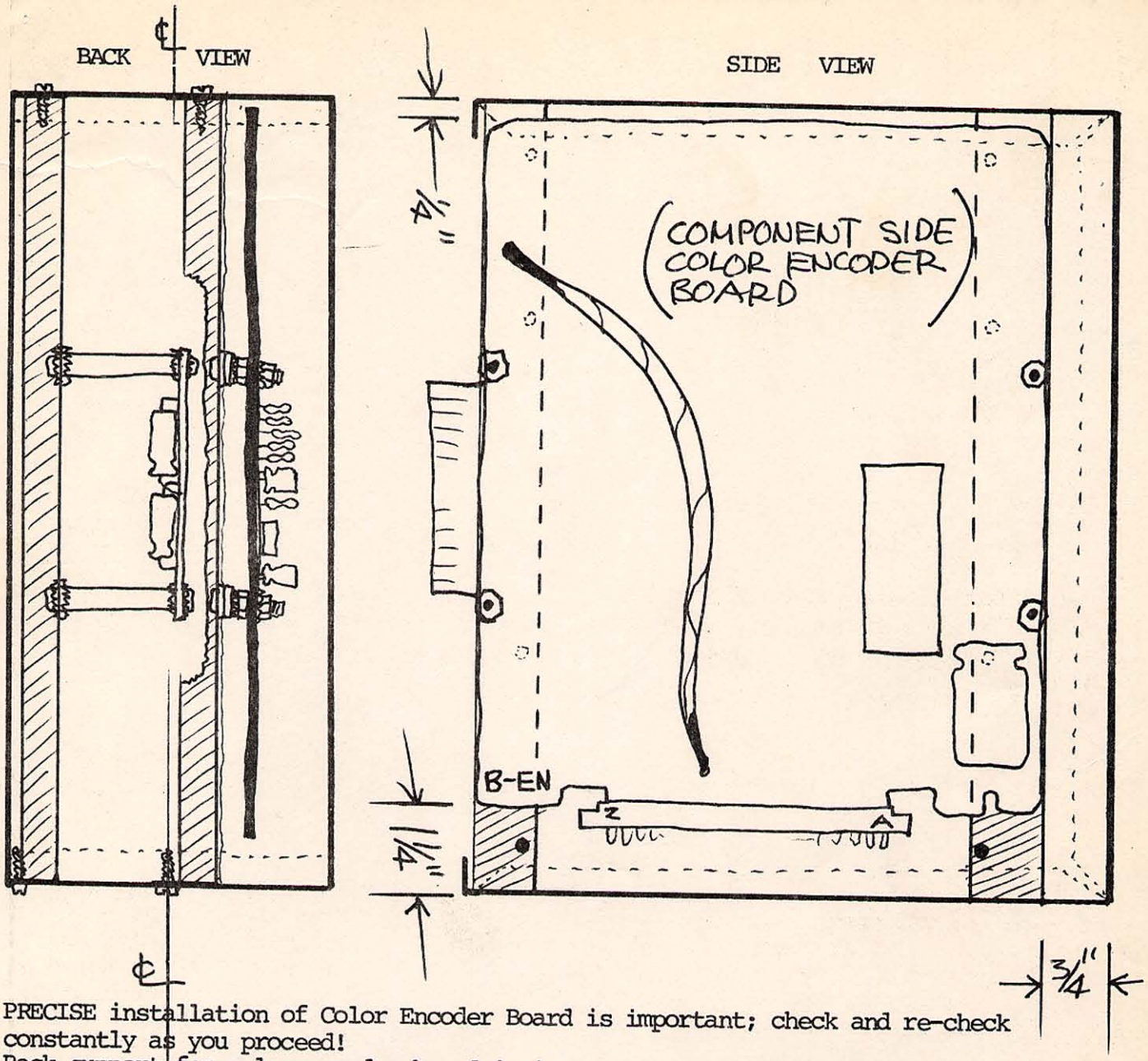
NOTE WELL

THIS DOCUMENTATION OF COLOR ENCODER IS OUT OF DATE

THE SONY ENCODER BOARD IS NO LONGER AVAILABLE

THE NEW DISIGN IN PROGRESS IS LIKELY??? TO BE VERY SIMILAR.

USE FOR GENERAL REFERENCE ONLY??



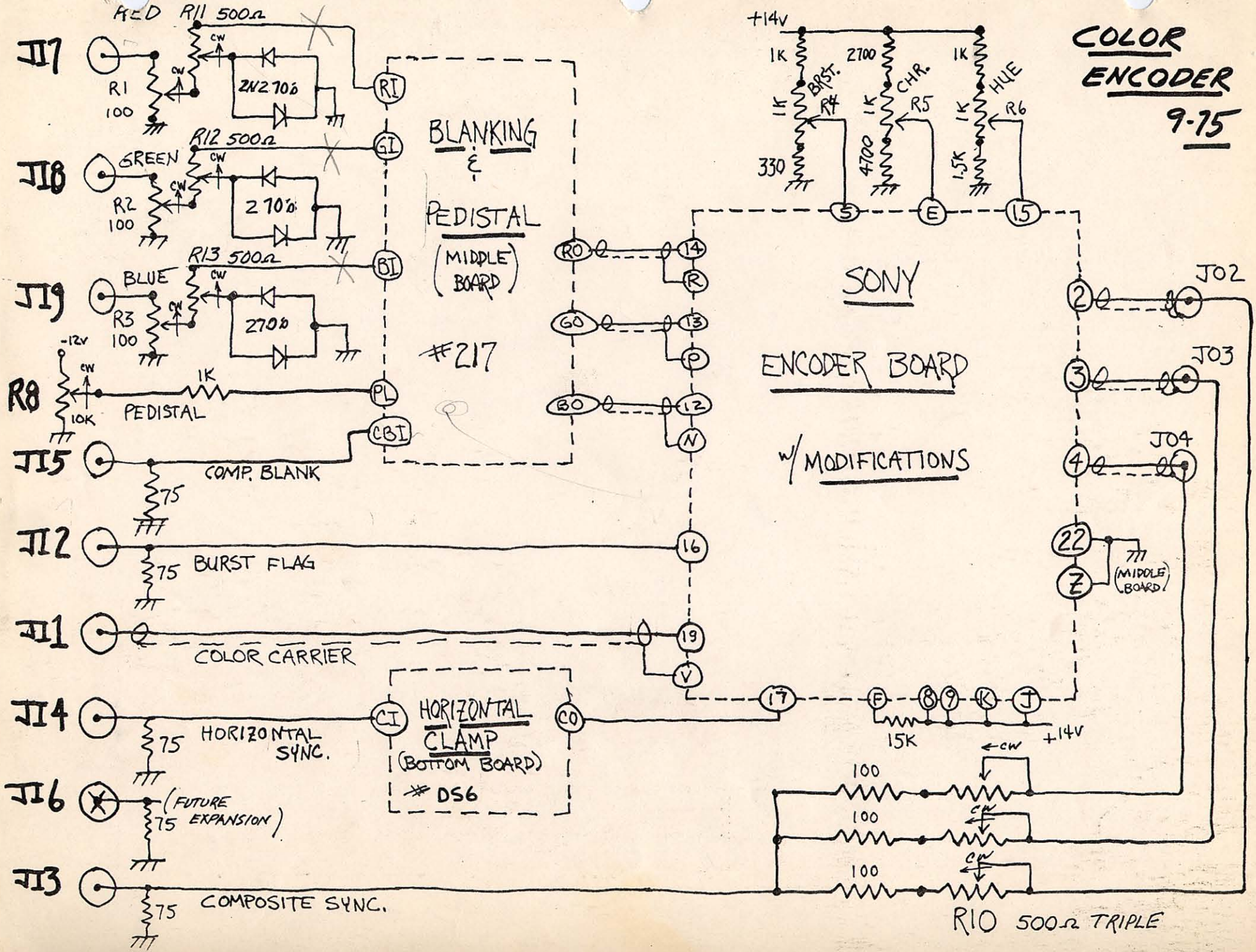
PRECISE installation of Color Encoder Board is important; check and re-check constantly as you proceed!
Back support for color encoder board is installed identical in position and orientation as is back support for the three printed circuit boards; see BACK VIEW pictorial. NOTE the position of bottom screw is on center-line of module and the top screw is abit shifted to right as usual. Be sure to cut plently of clearance in the support bracket for the middle printed circuit board to clear; see BACK VIEW pictorial.

Check for plently of clearance, made up of flat washers, star washer and nut, between encoder board and support bracket. As a 'safety measure' against board ever touching the supports due to bending etc, insulate the supports with a couple layers of super-tape or very thin rubber...

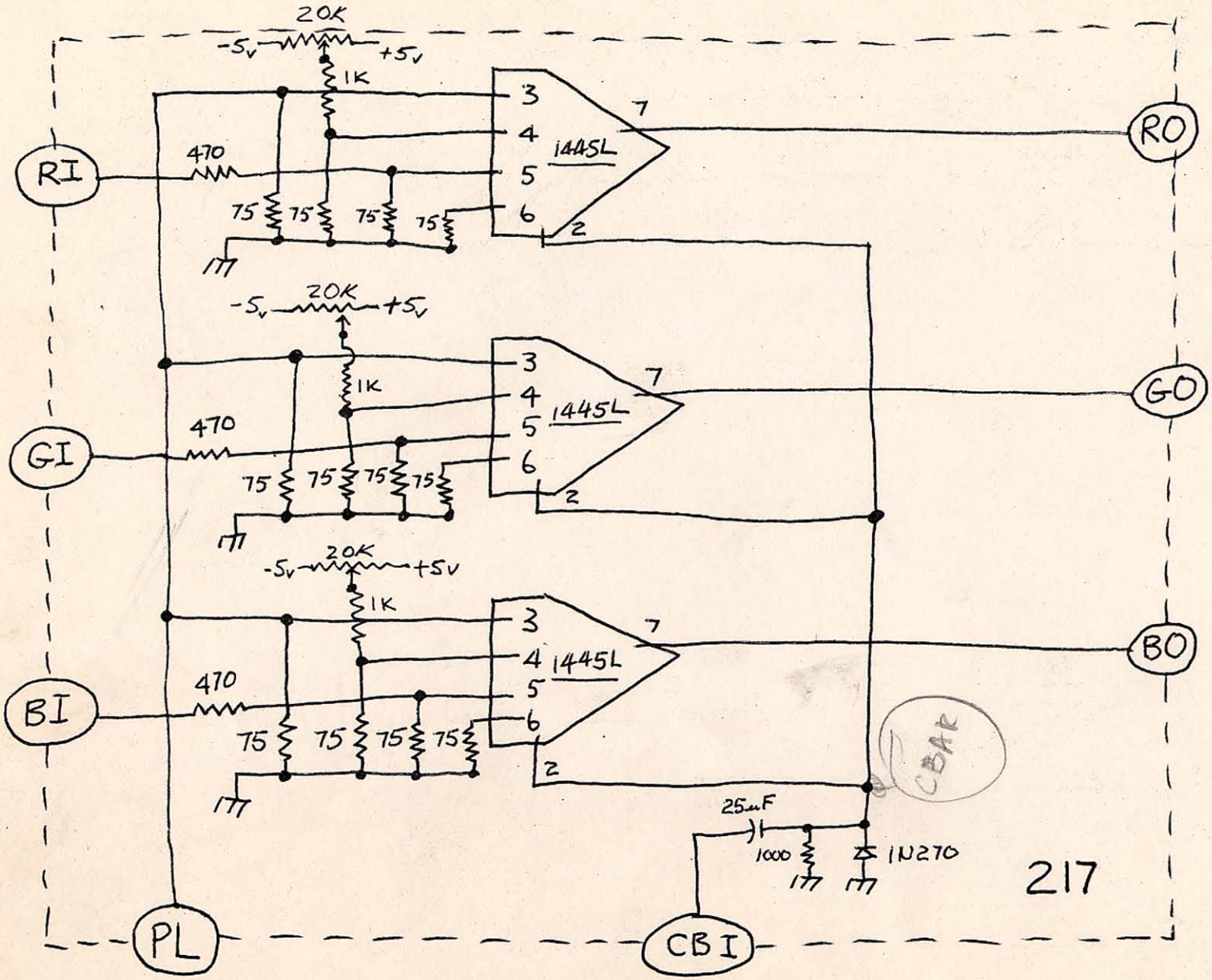
Front support bracket for color encoder board is installed upside-down and backwards to all other support brackets in module. This will put the screws for top-mounting of the bracket on center-line of module (not shown in pictorials). It is not necessary to cut away the lip of this front support bracket as done for the back support bracket.

NOTE the top 1/4", bottom 1 1/4" and front 3/4" clearances for encoder board from the module chassis; see SIDE VIEW pictorial.

You must drill 4 holes in encoder board for mounting it to the support brackets; be absolutley sure of correct allignment and positioning prior to drilling board.


COLOR ENCODER 9-75
SONY
ENCODER BOARD
W/ MODIFICATIONS
#217
BLANKING & PEDISTAL (MIDDLE BOARD)
HORIZONTAL CLAMP (BOTTOM BOARD) #DS6
RED
GREEN
BLUE
R11 500Ω
R12 500Ω
R13 500Ω
R1 100
R2 100
R3 100
2N2706
270Ω
-12V
1K
10K
PEDISTAL
COMP. BLANK
BURST FLAG
COLOR CARRIER
HORIZONTAL SYNC.
(FUTURE EXPANSION)
COMPOSITE SYNC.
75
+14V
1K
2700
330
4700
1.5K
BRST.
CHR.
HUE
R4
R5
R6
15K
100
R10 500Ω TRIPLE
J02
J03
J04
J7
J8
J9
J15
J12
J11
J14
J16
J13
R8
RI
GI
BI
PL
CBI
RO
GO
BO
CI
CO
CW

BLANKING & PEDISIAL BOARD
(MIDDLE BOARD)

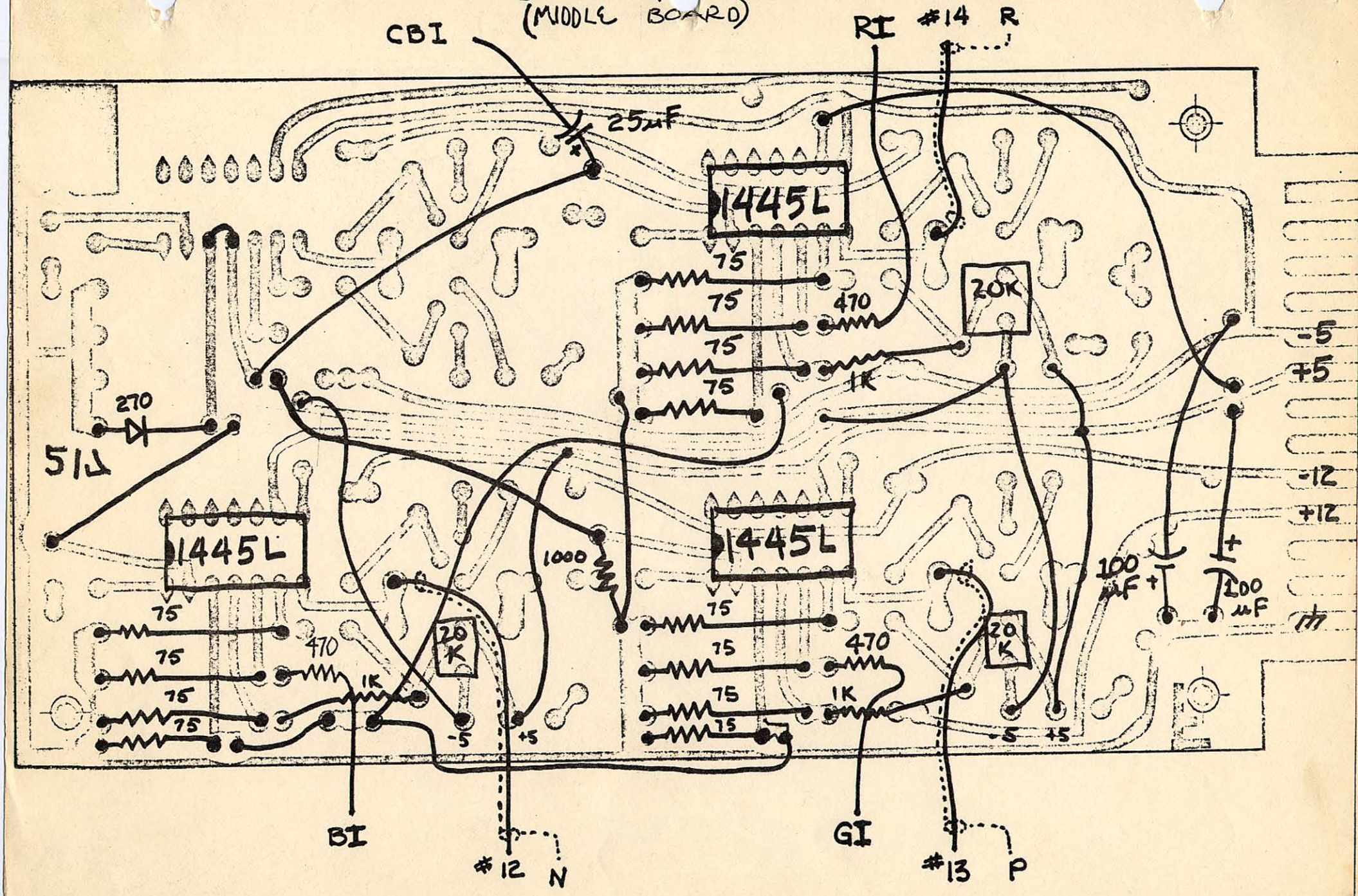


TRIMMERS ARE
RED PEDISTAL
GREEN PEDISTAL
BLUE PEDISTAL
ADJUST

217

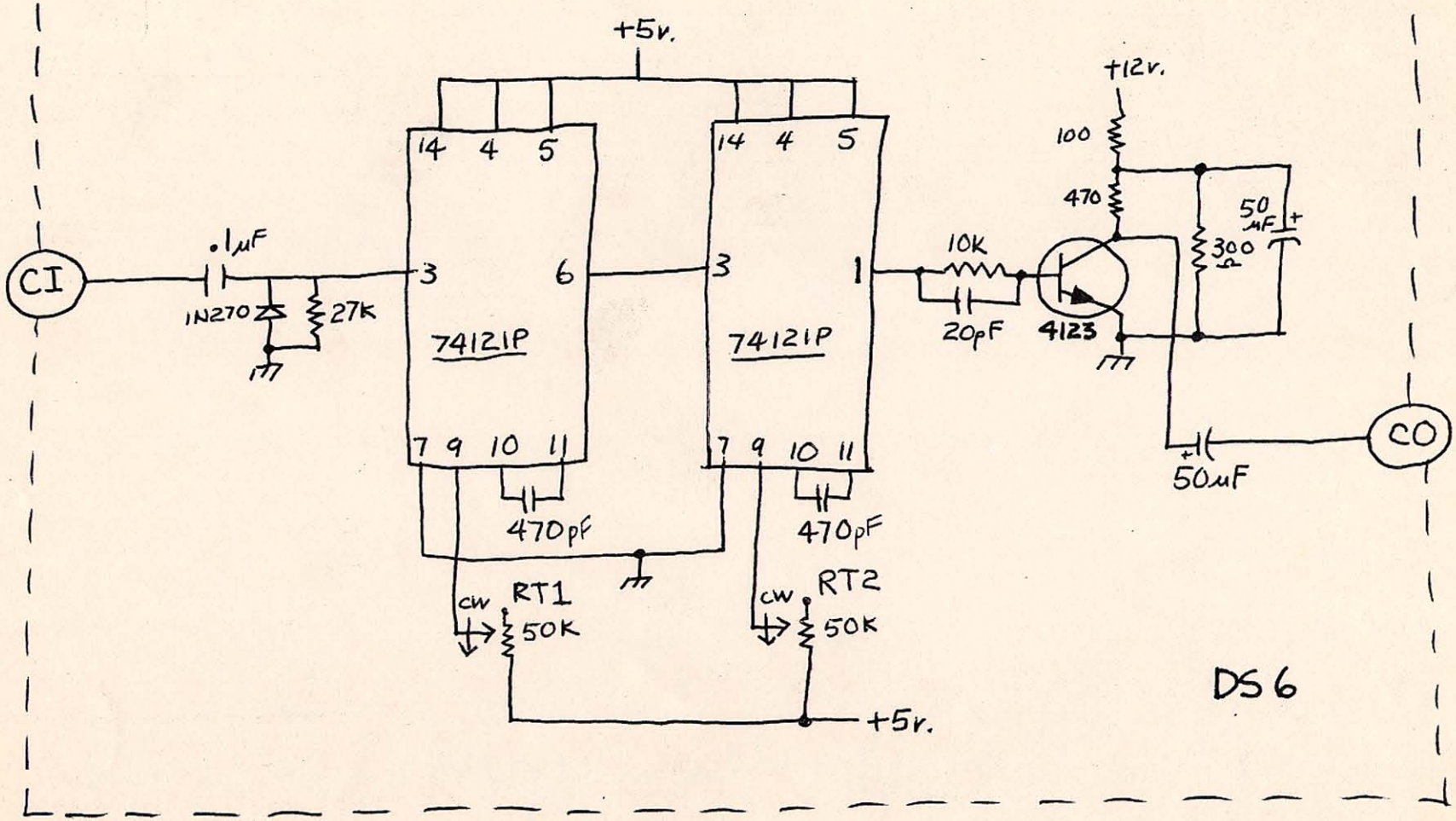
COLOR ENCODER

BLANKIN & PEDISTAL
(MIDDLE BOARD)

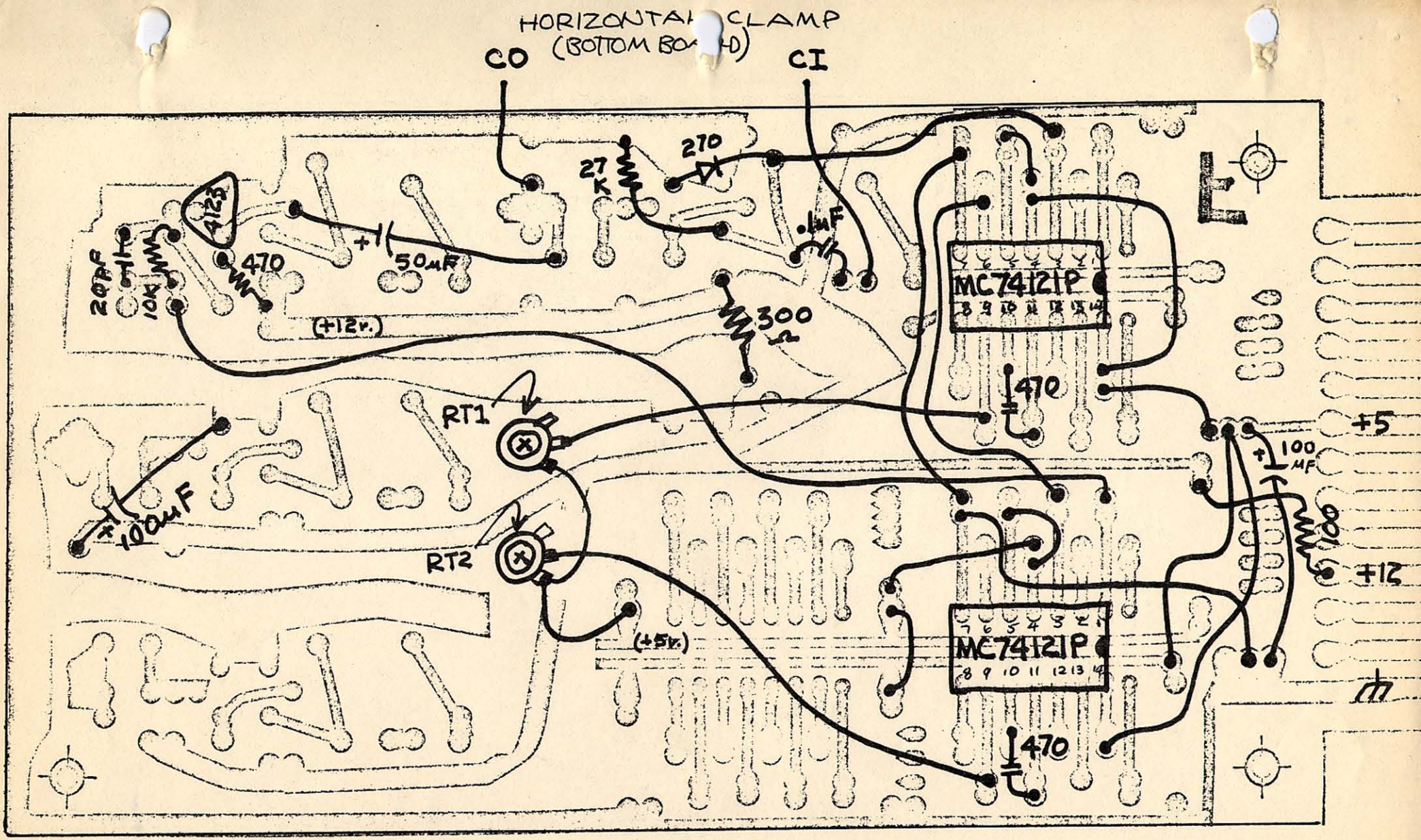


COLOR ENCODER

HORIZONTAL CLAMP BOARD
(BOTTOM BOARD)

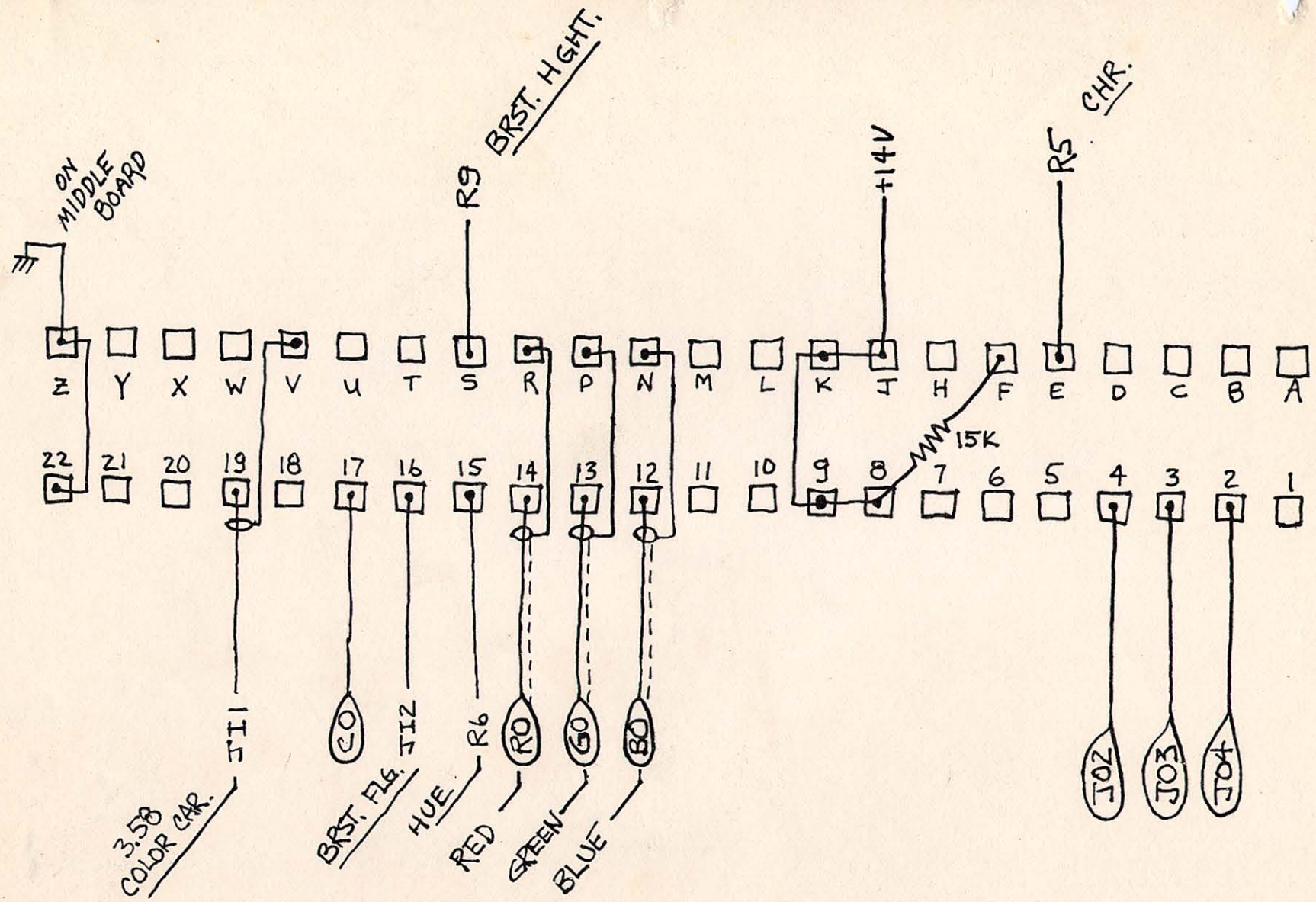


COLOR ENCODER



NOTE: "KNOTCH" ON MC7412IP (IC's); THEY ARE ATYPICAL TO OTHER IC INSTALLATIONS...
RT1 & RT2 SHOULD NOT TOUCH ANY PATHS...

COLOR ENCODER



22 thru 1 = FOIL SIDE
Z thru A = COMPONENT SIDE

COLOR ENCODER BOARD

44PIN - CONNECTOR DIAGRAM

update 9-75

21		525-1185	75 OHM, 1/4 W RES	.06	A	COL
1		525-1202	100 OHM, 1/4 W RES	.06	A	COL
1		525-1263	270 OHM, 1/4 W RES	.06	A	COL
2		525-1270	300 OHM, 1/4 W RES	.06	A	COL
1		525-1276	330 OHM, 1/4 W RES	.06	A	COL
4		525-1302	470 OHM, 1/4 W RES	.06	A	COL
7		525-1356	1KØ OHM, 1/4 W RES	.06	A	COL
1		525-1421	2.7KOHM, 1/4 W RES	.06	A	COL
1		525-1461	4.7KOHM, 1/4 W RES	.06	A	COL
1		525-1522	10KKOHM, 1/4 W RES	.06	A	COL
3		525-1550	15KKOHM, 1/4 W RES	.06	A	COL
1		525-1603	27KKOHM, 1/4 W RES	.06	A	COL
3	9F073	U1	100 OHM POT 1/4SFT	1.45	N	COL
3	10F683	501UC	500 TRIM,LOK,PNL-MT	3.00	N	COL
3	9F078	U4	1K OHM POT 1/4SF	1.45	N	COL
1	9F081	U6	2K OHM POT 1/4SFT	1.45	N	COL
1	10F468	103UA	10K OHM POT 1/4SFAB	2.21	N	COL
1	10F588		500 OHM POT 1/4SFTB	5.02	N	COL
2			50K TRIM		N	COL
3	12F9800	3389F	20K TRIM BD-MT	.65	N	COL
1	14F557	DM15-200J	20 PF,DIP-MICA CAP	.18	N	COL
2	14F592	DM15-417J	470 PF,DIP-MICA CAP	.32	N	COL
1	67F313	TDC-104Z	.1 MF,DSC-CER CAP.	.16	N	COL
1	710-1254	3-G-025	25 MF,25VDC,ELEC.	.46	A	COL
3	710-1260	7-G-1000	100 MF,25VDC,ELEC.	.24	A	COL
6	553-0914	1N914B	SIL-DIODE	.19	A	COL
8		1N270	GERM-DIODE	.22	S	COL
1		2N4123	NPN TRANS	.22	S	COL
3		MC1445L	DIP GAIN-CONT.AMP	1.90	S	COL
2		MC74121P	DIP MONO.VIB.	1.18	S	COL
1			217 P-C BOARD			COL
1			P-C BOARD,DS6			COL
1		CEN-4092-5	COLOR ENCODER BRD	400.00	AP	COL
1	713-6140	225-2222-401	44CONTACT CON	5.89	A	COL
12	39F1337	13-236	BNC,FM-CHS.MT.	.72	N	COL
1			CHASSIS,COL.ENC-FAC	8.25	DG	COL
6		RB-67-1-DC-M-L-9	KNOB,BLACK/INLAY29	1.00	R	COL
1		RB-67-1-DC-M-L-9	KNOB,BLACK/INLAY10	1.00	R	COL
1		RB-67-1-DC-M-L-9	KNOB,BLACK/INLAY12	1.00	R	COL
1		RB-67-1-DC-M-L-9	KNOB,BLACK/INLAY11	1.00	R	COL

NOTE WELL
THES PARTS ARE INCLUDED IN MASTER PARTS LIST

POWER SUPPLY

The power supplies are purchased modules and should come with complete documentation; if not request from LAMDA.

In the IP, power supply regulation and high frequency transient response are critical. Substitution of other power supply modules is NOT recommended.

In each box all corresponding terminals of the 10 pin Jones connector are connected together.

The output of the power supplies are connected to the appropriate pin of one of the connectors.

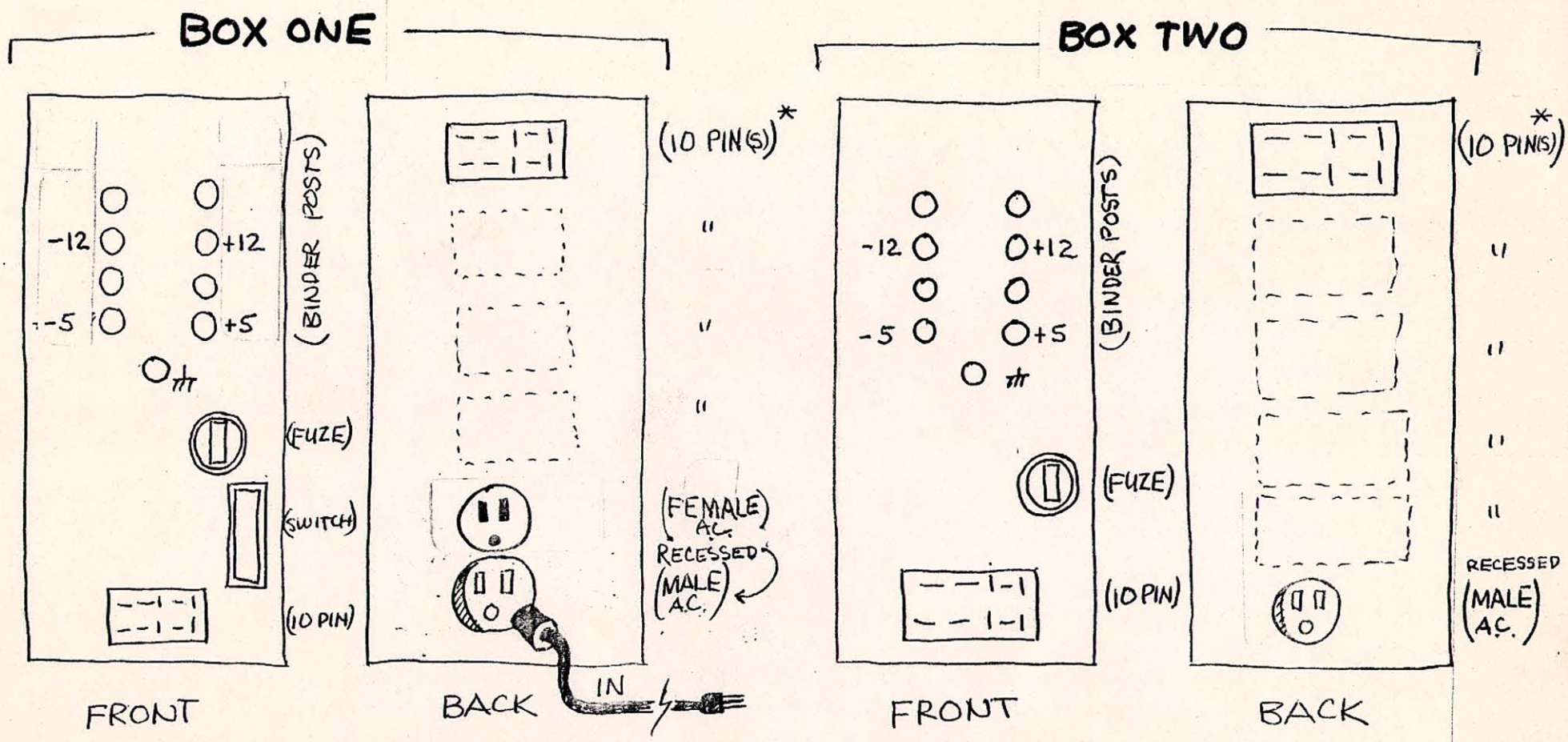
In both power supplies (Box one and Box two), the binding post terminals are connected to the appropriate 10 pin Jones.

A cable with two male Jones plugs and corresponding pins connected together is used to communicate power between the boxes.

One side of each box should be covered with perferated metal or screen to allow for ventilation. This side should never be blocked to prevent ventilation. DO NOT let transistors touch screen.

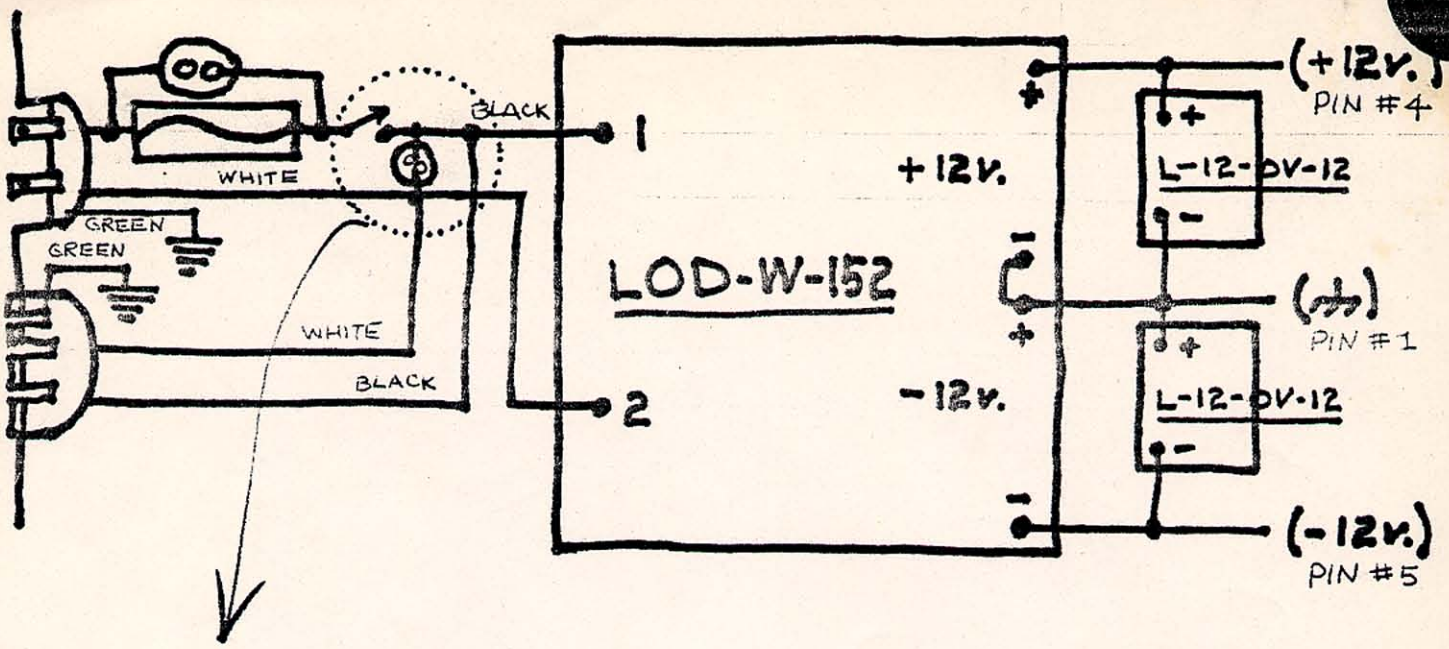
The 110 v. AC which powers the power supplies is the only potentially lethal voltage in the IP. BE CAREFUL AND WATCH YOUR FINGERS.

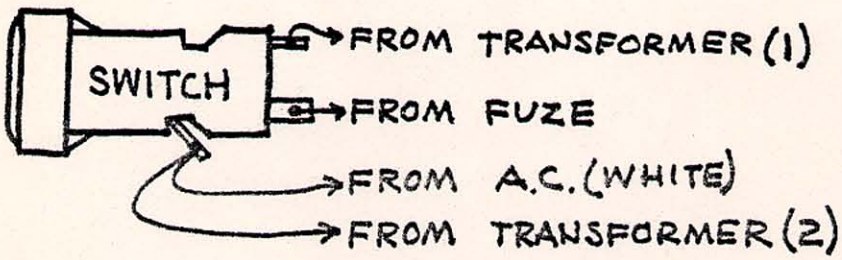
--Box one contains +12, -12 power supplies.
--Box two contains +5, -5 power supplies.



* TRY TO MOUNT AS MANY 10 PIN CINCH-JONES (FEMALES) AS POSSIBLE.

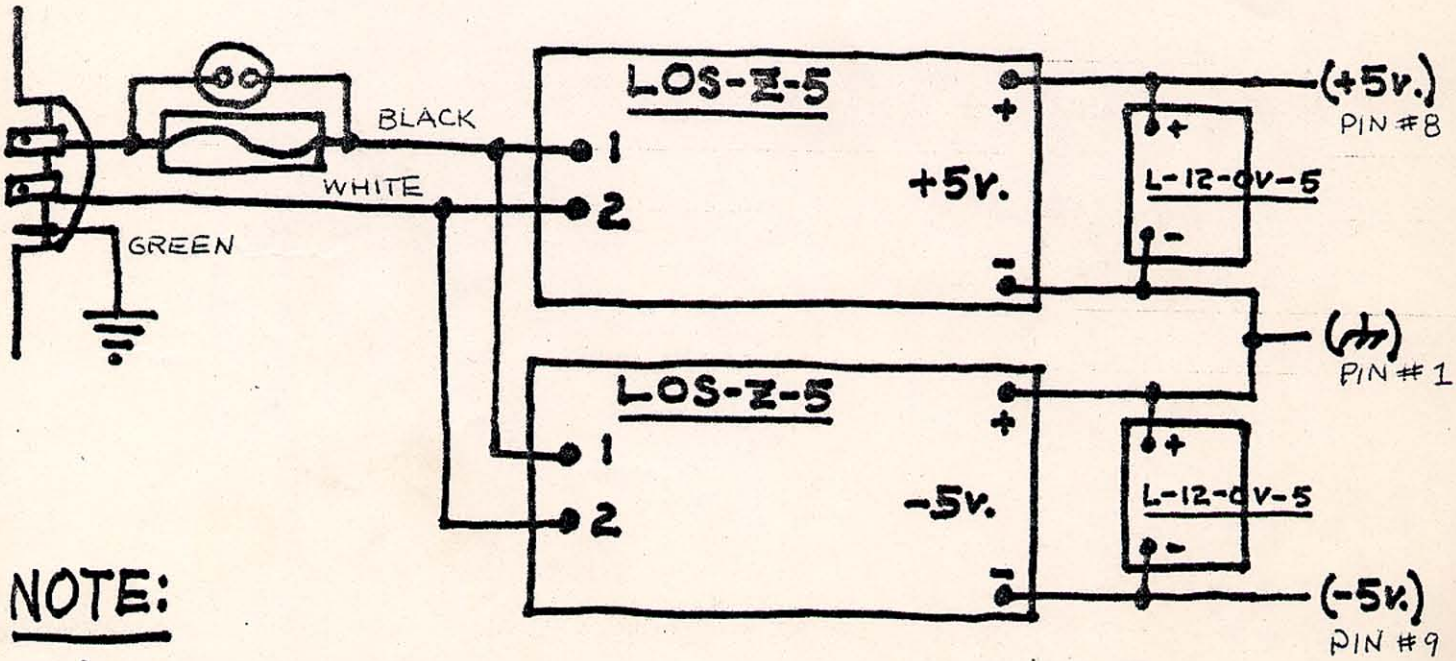
- A.C. POWER IS JUMPED FROM "BOX ONE" TO "BOX TWO" BY MALE-FEMALE A.C. CORD SO AS TO BE SWITCHED ON/OFF BY COMMON SWITCH ON 'POWER I'.
- "BOX ONE" AND "BOX TWO" ARE ALWAYS CONNECTED BY ONE MALE-MALE 10 PIN CABLE SO AS TO MAKE ALL 10 PIN CONNECTORS HAVE ALL POWER SUPPLY VOLTAGES.





BOX ONE

BOX TWO



NOTE:

⏚ - MEANS GROUND TO METAL BOX

⏚ - MEANS GROUND TO PIN #1

POWER1.IP 17-DEC-75 PAGE 1

1	LOL-W-152		12V,5AMP POWER SUP	87.00		L	PS1
2	L-12-OV-12		NON-ADJ. OVP	8.00		L	PS1
1	59F1740	102090	POWER SWITCH,AMBER	3.97	-2	N	PS1
1	27F784	344125A	110V LAMP-FUZE KNOB	4.06	-1	N	PS1
1	39F1232	160-4-N	AC RECP.FM*2POLE-GR	1.58	-2	N	PS1
1	39F1233	160-5-N	AC RECP.M*2POLE-GRD	2.06		N	PS1
5	39-F-255	S-310-FP	JONES FM*CHS.MT.	1.21	-6	N	PS1
1	39F1547	111-0103-001	POST BLACK 10	.52	-4	N	PS1
1	39F1552	111-0106-001	POST BROWN 1	.52		N	PS1
1	39F1546	111-0102-001	POST RED 2	.52	-4	N	PS1
1	39F1550	111-0105-001	POST ORANGE 3	.52		N	PS1
1	39F1551	111-0107-001	POST YELLOW 4	.52	-4	N	PS1
1	39F1548	111-0104-001	POST GREEN 5	.52	-4	N	PS1
1	39F1554	111-0110-001	POST BLUE 6	.52		N	PS1
1	39F1556	111-0112-001	POST VOILET 7	.52		N	PS1
	39F1557	111-0113-001	POST GRAY 8	.52		N	PS1
1	39F1545	111-0101-001	POST WHITE 9	.52		N	PS1
1			CHASSIS,POWER 1-FAC	8.25		DG	PS1

POWER2.IP 17-DEC-75 PAGE 1

~~1~~	~~[illegible]~~		~~[illegible]~~	~~[illegible]~~		~~L~~	~~PS2~~
2	LOS-Z-5		5V,5AMP POWER SUP	32.00		L	PS2
~~1~~	~~[illegible]~~		~~[illegible]~~	~~[illegible]~~		~~L~~	~~PS2~~
	L-12-OV-5		NON-ADJ. OVP	8.00		L	PS2
1	39F1547	111-0103-001	POST BLACK 10	.52		N	PS2
1	39F1552	111-0106-001	POST BROWN 1	.52		N	PS2
1	39F1546	111-0102-001	POST RED 2	.52		N	PS2
1	39F1550	111-0105-001	POST ORANGE 3	.52		N	PS2
1	39F1551	111-0107-001	POST YELLOW 4	.52		N	PS2
1	39F1548	111-0104-001	POST GREEN 5	.52		N	PS2
1	39F1554	111-0110-001	POST BLUE 6	.52		N	PS2
1	39F1556	111-0112-001	POST VOILET 7	.52		N	PS2
	39F1557	111-0113-001	POST GRAY 8	.52		N	PS2
1	39F1545	111-0101-001	POST WHITE 9	.52		N	PS2
1	39F1233	160-5-N	AC RECP.M*2POLE-GRD	2.06		N	PS2
1	27F784	344125A	110V LAMP-FUZE KNOB	4.06		N	PS2
5	39-F-255	S-310-FP	JONES FM*CHS.MT.	1.21		N	PS2
1			CHASSIS,POWER 2-FAC	8.25		DG	PS2

Electronic Visualization Laboratory
Department of Information Engineering and School of Art and Design
University of Illinois at Chicago Circle Box 4348 Chicago, Ill. 60680
Telephone: 312 996 4621

HI Z* LOW PASS INPUT MODUAL

This modual alows high inpedence sorces shuch as audio, and audio synthesizer controll voltages it be inputed to the I.P.

It is based on the input modual so use its part list etc.

The controles r1,r2 work in the same way as the camer input modual. except the idea is to supress some of the incoming signil. switch #s1 either ac cupples input (up) or dc cupples(down). switch #s2 selects lowpass filter. center is no filter up is maximum filtration, down is less filtration(averaging).

thanks to Mi Mi Shevitz for help documenting.

The tripil through ~~tmixm~~ dubble pole switch

newark # 61f867
or 61f1251
or 29f2284

J11

R1H

J02

J01

R2H

J04

J03

HORIZONTAL

FRONT FACE



J12

R1V

J02

J01

R2V

J04

J03

VERTICAL

SWEEP GENERATOR